

# LifeSim Childhood: Extrapolating Intervention Effects and Public Cost Savings from Birth to Adolescence in the UK

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**Abstract** Economic evaluation of early childhood interventions is challenging because it is hard to extrapolate the full range of long-term benefits and public cost savings from short-term effectiveness evidence. One way to address this issue is through the use of microsimulation. This paper introduces a closed birth cohort childhood microsimulation model, LifeSim Childhood, that is capable of extrapolating the long-term effects of various childhood circumstances on a broad range of health, educational, and social outcomes and public cost savings up to age 17 in the UK. We use regression-based long-jump microsimulation that simulates outcomes at each age directly from total-effect parameters estimated using bespoke longitudinal birth-cohort regressions, without tracing indirect mediating pathways. The aims of this paper are to describe the general modelling approach and methods underpinning LifeSim Childhood, to present a simple illustration of how the model can be used to simulate the effects of reducing early childhood poverty, and to compare our illustrative results with estimates from quasi-experimental studies. We use data from the Millennium Cohort Study which follows children born in the UK around the year 2000. We focus on exposure–outcome relationships that can be plausibly interpreted as causal based on interdisciplinary theory and evidence, and estimate total effects conditional on an explicitly justified set of confounding variables. As well as describing our data inputs, regression analysis methods and simulation methods, we apply the model to four hypothetical income-shifting scenarios in early childhood and estimate the general magnitude of long-term public cost savings and wellbeing benefits, alongside a battery of more specific outcomes.

DOI: <https://doi.org/10.34196/ijm.00343>

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**Author Keywords:** Childhood, Early Years, Microsimulation, Long Term Effects, Wellbeing, Public Costs

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# 1. Introduction

Early childhood circumstances from conception to age five have been shown to have important long-term effects on individual life chances across a range of domains including health, education, employment and crime (Heckman, 2012; Almond et al., 2018). However, it is challenging to estimate the magnitude of the full range of long-term life outcomes and associated public cost savings across varying policy domains. This is because the effectiveness data collected from trials, quasi-experiments and other intervention studies tends to focus on a narrow set of short-term policy targets, and follow-up data on long-term outcomes takes decades to accumulate (Skarda et al., 2022). Previous attempts at long-term early childhood policy modelling have tended to use special-purpose models that focus on a specific intervention (e.g. García et al., 2020) and cannot be re-used to compare the effects of other interventions. Milne and colleagues from the COMPASS research centre at the University of Auckland have developed a general-purpose childhood microsimulation model for New Zealand (Milne, 2015) called “Modelling the Early life-course (MELC)” that has recently been extended to age 24 as the “Better Start Model” (Milne et al., 2024). However, no such model yet exists for the UK and no previous model of this kind has been capable of simulating the general summary measure of wellbeing known as the “WELLBY” (a one point improvement in life satisfaction on a scale of 0 to 10 for one year), which is recommended by the UK Treasury as a general way of comparing the cost-effectiveness of different interventions across different policy sectors at different ages (Frijters and Krekel, 2021; Krekel and Layard, 2023). Microsimulation gives us the flexibility to create a re-useable general-purpose long-term model by combining multiple data sources that can be continuously updated.

We have previously started the process of developing a general long-term childhood policy model that can be re-used to evaluate different interventions, known as LifeSim (Skarda et al., 2021; Skarda et al., 2022). Our prototype model, which we call “LifeSim 2021”, was a discrete event simulation of annual outcomes from birth to death. However, from age 0 to 17 it was primarily based on simply carrying forward observations from longitudinal data, rather than explicit modelling, and so the main capability of LifeSim 2021 was extrapolating outcomes from late adolescence to late adulthood, rather than extrapolating early childhood circumstances to outcomes up to late adolescence. Consequently, here we present the first general-purpose UK childhood microsimulation model capable of extrapolating childhood circumstances to a rich set of outcomes in later childhood and adolescence, which we call LifeSim Childhood. This new model is capable of inputting policy effects on childhood circumstances (age 0 to 14) and outputting a wide range of extrapolated outcomes and public cost savings up to age 17, including WELLBYs. We refer to childhood circumstances as “exposures” in a causal inference context, and as “policy targets” in a policy context; we also sometimes refer to them as “risk factors” or “protective factors” when they are measured as a binary variable, like poverty, rather than an ordinal variable like income quintile group or a continuous variable like log income.

We use regression-based long-jump microsimulation that simulates outcomes at each age directly from total-effect parameters estimated using bespoke longitudinal birth-cohort regressions, without tracing indirect mediating pathways. This approach ensures that we estimate the total effects of all mediating pathways and that our findings are closely tied to the real long-term outcomes observed in longitudinal data. However, it is potentially less conservative than a conventional discrete event simulation approach that models transitions from one period to the next in a relay fashion, based only on a core sub-set of mediating pathways that are explicitly modelled. We focus on exposure-outcome relationships that can be plausibly interpreted as causal based on interdisciplinary theory and evidence, and estimate total effects conditional on an explicitly justified set of confounding variables. LifeSim Childhood is thus a tool for providing policy-relevant estimates of total long-term effect magnitudes, not for analysing mediating pathways or for providing new scientific evidence for or against the existence of specific causal links. The model builds on LifeSim 2021 by analysing the effects of early childhood circumstances on important adolescent outcomes known to have long-term effects on health and other life outcomes in adulthood (Villadsen et al., 2023). Like LifeSim 2021, however, it remains a closed birth cohort model that only estimates long-term effects for a single birth cohort of children and does not model changes to household structure or effects on parents and siblings.

The main purpose of this paper is to describe and illustrate the use of LifeSim Childhood. Using an increase in early childhood income as an example, we describe our model and causal inference strategy for estimating the magnitude of impacts, and illustrate how it can be used to extrapolate the consequences up to age 17. The policy target (early years income), the change to the policy target

(increasing income for a specific population) and the model used for estimation can be updated and changed as necessary for future research. Our model simulates various childhood outcomes that could subsequently be plugged into our adulthood model to estimate life-course outcomes, and we plan to do that and describe the methods and results in subsequent papers. Our model is a micro-economic evaluation tool that aims to predict the specific costs and benefits of specific interventions, rather than a macro-economic forecasting tool that aims to predict future trends in general population outcomes. Where possible, we compare our model simulated estimates to external experimental and quasi-experimental estimates from the literature, to ensure that our estimates are in the right ballpark.

To illustrate how the model can be used in practice, we follow [Villadsen et al. \(2023\)](#) in modelling the potential effect of four simple hypothetical income-increasing scenarios for groups of households split by their position in each fifth of the MCS household income distribution: (1) increasing the income of the poorest fifth to that of the second poorest fifth; (2) increasing the income of the two poorest fifths up to that of the middle fifth; (3) increasing the income of the poorest to the richest fifth, and (4) increasing everyone's income to that of the richest fifth. We report outcomes and costs by age and disaggregate costs by source (hospitalisation, disability, conduct disorder, special education needs (SEN), truancy and exclusion). These scenarios are not realistic policy options for specific tax-benefit reforms but are just simple illustrative examples asking what would happen if in theory it were possible to make perfectly targeted cash transfers with no administrative costs or deadweight losses due to the effects of tax-benefit reforms on economic behaviour.

We use Millennium Cohort Study (MCS) birth cohort data on about 15,380 babies born in the UK around 2000/2001 and followed up to age 17. We use multiple imputation to handle missing values due to attrition and non-response. We model a set of policy-relevant outcomes from age 3 to 17, including cognitive skills, socio-emotional and behavioural problems, educational attainment, smoking, obesity, self-reported health, psychological distress, wellbeing, and six cost-bearing outcomes. In this study, as an illustrative example, we focus on a single early years risk factor - mean household income age 0 to 5. However, the same approach can be applied to a wide range of other risk factors and circumstances during childhood and adolescence that policy makers often target for prevention, support services or both. We take an explicitly justified approach to causal inference based on the principles set out by [Pearl \(2009\)](#). We focus on estimating the total magnitude of causal effects that are already well-established or at least can be given a plausible justification on the basis of current inter-disciplinary scientific knowledge, and we only include confounders for which there is a plausible justification based on a clear set of causal inference rules. We make our causal inference assumptions explicit, we identify theory and evidence to support them, and we identify important uncertainties and controversies about our assumptions and how far this might materially impact our conclusions. We chose income as an illustrative risk factor as there have been many experimental and quasi-experimental studies of the long-term effects of early childhood income on specific later childhood outcomes ([Cooper and Stewart, 2021](#)). These are reviewed later in section 3.6 to provide evidence for the external validity of our estimates, derived from observation data, which control only for measured confounding factors.

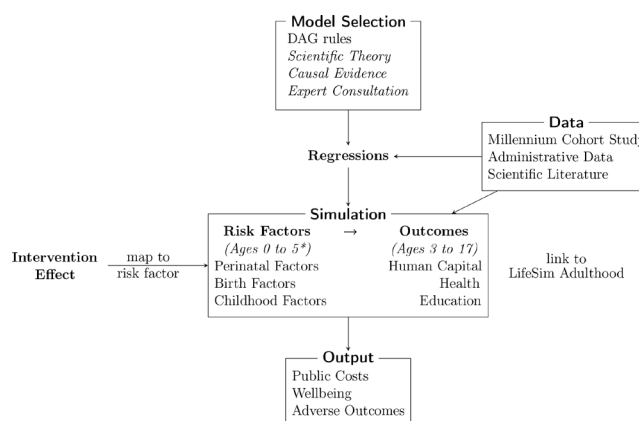
LifeSim Childhood is designed as a general modelling platform for three different policy-relevant purposes requiring estimation and comparison of long-term effect magnitudes. First, building the case for public investment in childhood, including the early years (conception to age 5), primary school years (age 5-10), and early adolescence (age 10-15). Second, comparing the long-term social costs and benefits of using policy interventions to modify different childhood circumstances. Third, assessing and comparing the value for money of different cross-sectoral childhood policy interventions. These are all extremely challenging tasks, and so LifeSim Childhood does not aim to be perfect. It merely aims to provide an embryonic methodology that can help to improve upon the existing information about the long-term consequences of childhood policies that is used in practice to inform decision-making, which at present consists of subjective opinions together with estimates based on extremely heterogeneous approaches that do not allow comparisons.

LifeSim Childhood makes a general background operating assumption that the relevant early childhood exposure has a stochastic causal effect on various policy-relevant outcomes in later childhood - including cognitive skills, emotional problems, behavioural problems, hospitalisation, special educational needs, school exclusion and educational attainment, among others. We use external theory and evidence to support the validity of this causal assumption and we use high-quality longitudinal observational data to provide a standardised and comparable way

of estimating the magnitudes of the total causal effects. The causal assumptions underpinning LifeSim Childhood are more securely grounded in external theory and evidence for some exposures and some outcomes than others, and the population average total effect magnitude is likely to depend on contextual factors including the nature and duration of change in the exposure. In the case of our illustrative example regarding early childhood income, however, we believe that our causal assumptions are securely grounded in line with the most recent and comprehensive reviews of evidence (*Candelora et al., 2025; Cooper and Stewart, 2021; Department for Work and Pensions, 2025; Page, 2024*), a large body of scientific theory from life course epidemiology (*Wagner et al., 2024*), developmental psychology (*Hertzman and Boyce, 2010*) and the economics of parental investment in childhood skills formation (*Francesconi and Heckman, 2016*). Children in low-income households face higher risks of poor physical and mental health, lower educational attainment, greater involvement with social care and justice systems and reduced employment and earnings in adulthood (*Marmot et al., 2010; Lai et al., 2019*). These higher risks are not entirely due to unobserved differences between rich and poor families. Although some studies find no statistically significant effects, the balance of experimental and quasi experimental evidence suggests that early childhood income has a non-zero causal effect on numerous developmental outcomes in later childhood, mediated through factors such as maternal mental health, parenting, and home environment (*Cooper and Stewart, 2021; Hill et al., 2013; Page, 2024; Department for Work and Pensions, 2025*). However, considerable uncertainty remains about the magnitude of these long-term effects on children's outcomes and whether the costs of reducing early childhood poverty are worth the benefits, compared with other ways of using public money. Our illustrative example aims to provide a reasonable empirical estimate of the population average magnitude of these long-term total effects, based on high-quality longitudinal observational data and controlling for a rich set of observed confounding variables. We do not start from a blank sheet and aim to prove beyond all possible doubt that these causal effects are non-zero, since that is not possible to do using observational data. Rather, we start from a holistic assessment of existing theory and evidence about the causal effects of early childhood income on later childhood outcomes and then use high-quality longitudinal observational data to provide a consistent and comparable way of estimating empirical magnitudes.

## 2. Methods

An overview of the structure of LifeSim childhood is presented in **Figure 1**. Modelling using LifeSim begins with the data drawn from the Millennium Cohort Study (MCS) from which we identify "risk factors" (ages 0 to 14) and "outcomes" (ages 3 to 17). In this sense, "risk factors" represent any early years variable that could be a potential policy intervention target for improving the "outcomes". Our current list of risk factors and outcomes has been chosen through consultation with an interdisciplinary group of academics, policy experts, stakeholders and policy makers.<sup>1</sup>



**Figure 1.** Structure of LifeSim Childhood

1. A non-exhaustive list of those consulted as part of our advisory group are included in appendix section A.9

Ongoing work will extend this to include risk factors up to age 14. The current list of risk factors includes early years household income, having a teenage mother, low birth weight, disability, delayed school readiness, etc. Outcomes are estimated from the age of the risk factor up to age 17, and will be extended to age 23 when the new sweep of the MCS is available. Our current outcomes include cognitive ability, socio-emotional scores, hospitalisation, exclusion from school, etc. The important adverse outcomes at age 17 we currently focus on are poor GCSE performance, psychological distress, obesity, smoking and poor health.

We make explicitly justified assumptions about causal inference based on principles set out by *Pearl (2009)*, and follow *Squires et al. (2016)* in taking a systematic approach to developing those assumptions and selecting appropriate models for estimating the magnitude of causal effects of risk factors on outcomes, based on review of the scientific literature and iterative consultation with stakeholders, experts, and members of the team. Through this process we have set up a set of causal inference rules to put together models of the effect of risk factors on each outcome based on directed acyclic graphs (DAGs).<sup>2</sup> We then use regression analyses run on the MCS data to estimate the relationships.

For the simulation we extract not only coefficients but also their standard errors and the distribution of residuals from the regressions to parametrise uncertainty. These are used as parameters in our simulation of the outcomes. We attach unit costs to various cost-bearing outcomes to estimate public costs. We use a parent-reported emotional symptoms score as a proxy indicator of life satisfaction - one domain of the parent-reported strengths and difficulties questionnaire (SDQ) - with sensitivity analysis using a two-item version that includes a peer problems domain score as well. This allows us to calculate public costs, subjective wellbeing and other life outcomes from age 3 to 17.<sup>3</sup>

Short term intervention effects from external studies, (e.g. improvements to birth weight or reduction in socio-emotional problems) or policy changes (e.g. increases in income due to changes to the tax benefit system or direct money transfer) can be input into the model to simulate long-term effects. If the exact risk factor is not available, the effects can be mapped onto an equivalent risk factor available in the MCS and input into the model.

## 2.1 Data

The main data source used in LifeSim childhood is the Millennium Cohort Study (MCS), a longitudinal cohort study of around 19,000 children born in the UK around the year 2000. The MCS surveys were conducted when the children were 9 months, 3 years, 5 years, 7 years, 11 years, 14 years, and 17 years.<sup>4</sup> After excluding twins and triplets and limiting our sample to those who responded to the age 3 survey, the final analytical sample consisted of 15,380 cohort members.<sup>5</sup> The response rate for the MCS cohort declined over time, affecting the availability of outcomes measured at later ages. To deal with missing data we use multiple imputation using chained equations. We work with 30 sets of the multiply imputed data for the 15,380 children which we use for the regression analysis and simulation.<sup>6</sup>

The variables we use for our estimation from the MCS can be roughly divided into three categories: risk factors, confounders, and outcomes.<sup>7</sup>

### 2.1.1 Policy Targets (Risk Factors)

The list of policy targets whose long-term effects could be estimated in LifeSim Childhood is broad and can be expanded based on availability in the MCS. This can include various maternal characteristics (e.g. teenage parent, maternal education, maternal mental health), household characteristics (e.g. income, single parent family), and child outcomes observed at birth (e.g. low birth weight, pre-term birth) and later in childhood (e.g. disability, school readiness, socio-emotional development).

Our illustrative example uses household income in early childhood as the policy target, based on the average OECD equivalised weekly income of the household measured when the child is 9 months, 3 years, and 5 years. We use this pooled measure to capture the household's permanent income

2. These rules are described in detail with examples in appendix section A.6.

3. We assume our risk factor "permanent income" affects outcomes from birth but we only model public costs and wellbeing gains from age three to age 17 in an effort to be conservative.

4. The study is ongoing and the next sweep of data from when the cohort members are around 23 was released in February 2026

5. 3,705 children are dropped due to non response in the second sweep at age 3 and 396 children who are twins or triplets are dropped. Twin and triplet households are dropped entirely.

6. The imputation process is described in more detail in appendix section A.5

7. In this paper the three groups of variables will be mutually exclusive but this is not the case for LifeSim in general. Appendix section A.1 defines these terms in more detail.

during early childhood. This avoids the risk of bias due to temporary income shocks that would occur if we focused only on income measured in one specific sweep of the survey. Our results specifically look at the effect of increasing income for families, thereby moving them up the income distribution<sup>8</sup> We assume that the timing of change to this “permanent income” starts prior to the point of measurement when the child is 9-months old and continues throughout early childhood to at least the age of five.

### 2.1.2 Confounders

The MCS data includes detailed information about the child, their parents and household at each sweep of the survey. Any of this information can be included as a potential confounder, but the final selection for each model is based on our DAG rules and consultation with experts.

For example, in our illustrative example with income as the policy target we use a set of eight confounders in our preferred model. We include basic demographic characteristics at age 9 months which we assume to be the same as those at birth: country, region within England, and ethnicity of the child. We also include some maternal characteristics: mother’s age at birth of the child, and maternal smoking during pregnancy. Finally we include some household characteristics: education of parents, parental disability and having a single parent.<sup>9</sup>

### 2.1.3 Outcomes

LifeSim Childhood can simulate the effect of a risk factor on a broad range of outcomes, limited only by data availability. However, in this paper we focus on three broad outcome categories, important adverse outcomes, wellbeing, and outcomes with direct public costs.

#### Adverse Outcomes

We simulate a set of five adverse health and educational outcomes measured in the MCS at age 17. These outcomes are all both important in their own right and also predictive of poor outcomes throughout adulthood (*Villadsen et al., 2023; Hale et al., 2015; Patton et al., 2016; Akasaki et al., 2019; Ploubidis et al., 2021; Gondek et al., 2021; Berg et al., 2022*).

The first, poor GCSEs, is a binary measure for poor self reported academic performance at the end of secondary school.<sup>10</sup> Specifically poor GCSEs is defined as not having 5 or more GCSEs,<sup>11</sup> including Maths and English, graded C or above.

The second adverse outcome is psychological distress determined by self reported Kessler score (K6). This is a self-report measure based on a 6 item questionnaire about anxiety and depression. Each response is scored from 1-4 and a score of 13 or over is considered to represent the screening threshold for probable clinical levels of psychological distress (*Kessler et al., 2003*).

The third adverse outcome is obesity based on a respondent meeting the UK90 growth reference chart obesity threshold for their age and sex at the time of interview (*Freeman et al., 1995*).

The fourth adverse outcome is ‘regular smoker’; an indicator for regular cigarette smoking based on self reported smoking of more than six cigarettes per week.

The final adverse outcome is poor/fair health an indicator for poor self reported health. Respondents are asked “How would you describe your health generally?” and pick a response of ‘excellent’, ‘very good’, ‘good’, ‘fair’ and ‘poor’. A response of ‘poor’ or ‘fair’ is considered as representing a report of being in poor health.

#### Wellbeing

Subjective wellbeing is an important policy outcome, which can be converted into the “wellbeing-year-point” (WELLBY) summary unit of benefit recommended by the UK Treasury, based on a one point improvement in life satisfaction for one year, valued at about £13,000 (*MacLennan et al., 2021*).

8. Income fifths are based on the distribution of household income within the multiply-imputed MCS data with age 3 UK population weights.

9. The confounders are described in more detail in section 2.3.1 and appendix section A.2.2 (see also **Table A3**).

10. General Certificate of Secondary Education (GCSE) results in England, Wales, and Northern Ireland, and National 5 (N5) results in Scotland

11. or 4 or more N5 results in Scotland graded D or above

We use the parent-reported Strength and Difficulties Questionnaire (SDQ) emotional symptoms scale from ages 3 to 17,<sup>12</sup> as an imperfect proxy indicator for life satisfaction as SDQ scores are consistently available throughout childhood in the MCS. The 0-10 scale for SDQ emotional symptoms (SDQES) is converted to a life satisfaction score (0-10) between 2 and 10 using a simple linear mapping. We used a conservative assumption-based affine transformation to convert SDQES to the life satisfaction scale:

$$LS = 2 + (10 - SDQES) \times \frac{8}{11} \quad (1)$$

Which simplifies to:

$$LS = 9.273 - 0.727 \times SDQES \quad (2)$$

Which yields mapped LS values ranging from 2 (for SDQES of 10) to 9.27 (for SDQES of 0).

A full range mapping based on matching the upper and lower bounds of the two scales would be:

$$LS = 10 - SDQES \quad (3)$$

This simple inversion of the SDQES scale would yield integer values of mapped LS from 0 to 10. We used an affine transformation based on two more conservative assumptions which reduce the scale of estimated WELLBY effects. We chose to map the higher bound SDQES value of 10 to a LS value of 2 rather than 0 to avoid generating unrealistically high rates of LS scores of 0 and 1, which for WELLBY valuation purposes HM Treasury treats as at or close to the death-equivalent anchor point. And we also used 11 as the denominator, rather than 10, because there are 11 discrete LS values from 0 to 10 and this gives a highest mapped LS value of 9.273 that stops just short of the highest possible LS value of 10. Both assumptions are conservative, in the sense that they both reduce the slope of the conversion factor and thereby reduce the scale of estimated WELLBY effects. Our assumed slope of -0.727 is about 27% smaller than the slope of -1 implied by a full range conversion, resulting in a 27% smaller mapped LS effect per estimated unit change in SDQES. Using this simple 0-10 scale allows us to directly translate our results to WELLBYs.

## Outcomes with Public Costs

We were able to capture many, though not all, of the childhood public costs that are potentially modifiable with early intervention using six outcomes available in the MCS: hospitalisation, disability, conduct disorder, special education needs (SEN), truancy and exclusion. In the following section, we describe each cost-bearing outcome, the costs in 2023 £s adjusted using the GDP deflator, the source of annual unit costs, and any other adjustments and assumptions made.<sup>13</sup>

Hospitalisation is based on parent report of admission to a hospital at ages 3, 5, 7, 11, 14 and 17 years. We do not distinguish between "avoidable" and "unavoidable" hospitalisation, so the baseline costs captured here should not be interpreted as the "costs of late intervention" as usually defined in terms of the incremental costs associated with children experiencing significant challenges in life compared with the standard health care costs needed for all children (**Chowdry and Fitzsimons, 2016**). Instead, we rely on our estimates to tell us how many instances of any hospitalisation are prevented by a change in the specific risk factor. We therefore rely on our causal inference modelling to estimate effects on public costs, rather than attempting to do this by distinguishing "avoidable" and "unavoidable" categories of public expenditure. Since MCS does not specify the details around recorded hospital admissions we use the average cost of a single inpatient hospitalisation for a child (age <18) in the UK of £1,587 (**Dale et al., 2024**). We do not capture all healthcare costs using this outcome and cost measure, just inpatient costs and not costs of primary care, outpatient care, and community care utilisation which are not possible reliably to estimate in the publicly available MCS data. However, inpatient costs make up a much larger proportion of preventable healthcare costs than primary, outpatient and community care costs (**Chowdry and Fitzsimons, 2016**).

12. We provide more information about using this measure in appendix section A.4. We also present simulation estimates with SDQ internalising - the sum of emotional and peer problem scores - as the life satisfaction measure instead in Appendix **Table A7**.

13. **Table A1** and **A2** in the appendix summarise the details of the costs and their sources.

Disability, based on parent reported long-term illness that affects the child's daily activity, is available at ages 3, 5, 7, 11, 14 and 17 years. We do not have good cost estimates for the disability in the UK. We limit ourselves to the health costs associated with disability. We use a cost of £1,008 per year per child to the NHS, the average annual cost of healthcare for a child in the UK (*Kelly et al., 2018*). This is likely an underestimate of the total cost of disability.

'Conduct Disorder' classification is based on the scores derived from the parent reported version of the SDQ conduct problems scale at ages 5, 7, 11, 14 and 17 years.<sup>14</sup> We follow *Skarda et al. (2021)*; *Goodman et al. (2003)*; and *Goodman et al. (2000)* in using a simple algorithm to estimate the probability of conduct disorder for each individual using their SDQ conduct problems scale. Age specific costs for conduct disorder are derived from *Bonin et al. (2011)*. We use a unit cost (annual cost per child with conduct disorder) of £3,092 between ages 5 and 10, £1,963 between ages 11 to 16, and £236 for age 17 and beyond. These costs decline as the child gets older as they only capture costs to the NHS, Social Services and Department of Education. The costs to the justice system which may increase with age are not currently included in the model.

Special Education Needs (SEN) is based on teacher/parent report of the child having a statement of special education needs. Costs for students with SEN statement are based on calculations for the average costs for students with an Education, Health and Care Plan (EHCP) for children in schools in 2023 amounting to an additional cost of £25,500 per year per EHCP student over the cost of a non EHCP student.<sup>15</sup> The cost was based on analysis by the Department for Education of costs associated with around 400,000 pupils with EHCPs in mainstream schools, special schools and independent special schools in 2023. Costs per EHCP pupil in 2023 are comparable to costs per SEN pupil in the early 2010s, since SEN statements were gradually phased out and replaced by EHCP plans during the early 2010s. The full (non-incremental) cost breaks down as £25,000 for place and top-up funding per EHCP, £2,000 for additional costs (e.g. SEN support services, support for inclusion, and therapies), £2,500 for additional cost of transport per EHCP (cost additional to expenditure on mainstream transport), and further administration and education psychology costs of £1,500 (*Department for Education, 2024a*; *Department for Education, 2024b*; *Acton et al., 2024*).

Truancy in the MCS is measured by parent reported absence from school at ages 11 and 14 years. A binary indicator for any truancy is used for the simulation. We cost truancy using the average cost to the Department of Education from regular truancy per child who is regularly absent from school, which is £943 per year per child (*Brookes et al., 2007*). The parents in the MCS are also asked about number of weeks of absence, a response of more than 5 weeks a year is considered regular truancy. We assume there is no cost to the public for any truancy that is not regular truancy.

Exclusion in the MCS is measured by parent reported exclusion from school ages 11 and 14 years. Parents are also asked about permanent exclusion at age 11. We cost exclusion using the cost of alternate provision, an alternative school for permanently excluded children to complete their schooling. Consequently, permanent exclusion is costed at £21,848 per year per child in alternative provision (*Bryant et al., 2018*). We assume no cost to the public for temporary exclusion.

## 2.2 Simulation Structure

The simulation in LifeSim Childhood runs a cohort of children picked from the MCS through the course of childhood several times or across several "universes" and produces the outcomes for each individual in the cohort in every universe. We perform a probabilistic sensitivity analysis using universes by introducing an element of randomness to the models built using the MCS.

We use the MCS data for the baseline characteristics of the cohort we run through the simulation. We use regression-based long-jump simulation which goes directly from the exposure to the outcome in each subsequent period without tracing out the mediating pathways at individual level (e.g. directly from age 3 to age 17). This is different from many other microsimulation methods such as the discrete event simulation approach we previously used in (*Skarda et al., 2021*) (the prior version of LifeSim) which use a "relay" approach, modelling outcomes sequentially period by period and tracing out the pathways at individual level. Both the long-jump and relay approaches to simulation have their

14. SDQ scores are also available at age 3 but we do not have a cost at that age and the algorithm we use to identify conduct disorder is not appropriately calibrated for this age group so we omit these outcomes.

15. The average cost per EHCP student is estimated at around £31,000 and the average cost for a non-EHCP student is estimated at around £5,500 in 2023.

benefits and downsides. The main difference is that the relay approach traces out full individual level causal pathways including mediating effects, whereas the long-jump approach only estimates total effects. The long-jump approach thus captures the total effect of all mediating pathways, without explicitly modelling those pathways, whereas the relay approach only captures the effects of mediating pathways that are explicitly modelled. The long-jump approach is potentially less conservative than the relay approach, for reasons explained later in the discussion section, but also more closely tied to the real long-term outcomes observed in the underpinning longitudinal data and potentially more comprehensive in the sense that it captures pathways that cannot be explicitly modelled using the relay approach.

We also prefer the long-jump modelling approach for LifeSim Childhood as a way to simplify the models to be estimated and minimise possible bias due to propagating bias and error from one period to the next. Since each effect is measured with some error/bias the error in the simulation when using the relay approach will likely increase with the number of years modelled. The long-jump simulation approach helps guard against this error being propagated and compounded through the model giving us estimates that are less likely to be biased.

In order to estimate the parameters for our long-jump modelling approach we run regressions estimating the effect of each risk factor on each outcome in each simulated year. We use linear regressions for cognitive ability (normal distribution), negative binomial regressions for SDQ scores and Kessler (count variables), and logistic regressions for the binary outcomes (Hospitalisation, disability, special education needs, truancy, exclusion, poor GCSEs, obesity, regular smoking, poor health). The point estimates as well as the variance of the estimates are extracted from the regression for use in the simulation. To get a nationally representative sample for estimation we use weights included in the MCS dataset, they adjust for attrition between the initial survey and the second sweep and the MCS's sampling strategy.

The simulation is not deterministic, rather parameter uncertainty is captured in the model in the following two ways. First, the coefficients used to simulate the outcomes are randomly drawn from a normal distribution with a mean equal to the point estimate and standard deviation equal to the standard error from the estimates obtained. These coefficients are not restricted to being positive or negative and are randomly drawn from the full distribution. Second, an error term is added based on a random draw from the distribution of residuals to each estimate. Doing this allows us to capture some element of the variance of the outcomes. Additionally, in the case of binary outcomes, our estimates allow us to come up with a predicted probability of the outcome for each individual, we predict success for each individual by comparing the predicted probability with an individual random draw from a uniform distribution between 0 and 1.

## 2.3 Estimation Strategy

In order to estimate the total effect magnitude in the absence of experimental or quasi-experimental data we follow **Pearl (2009)** in using directed acyclic graphs (DAGs) to specify the relevant causal pathways and guide choices about the selection of control variables. This allows us to bring together prior scientific knowledge to determine causal links between risk factors and outcomes in our data.<sup>16</sup> The depth of information about the individuals and parents in the MCS allows us to capture a rich set of confounders in the relationship between our risk factors and outcomes using a set of model selection rules, so our models are credible, conservative, and parsimonious.<sup>17</sup>

The main limitation of MCS in terms of relevant confounders is the absence of genetic variables, biomarkers, and direct measures of some early childhood adverse experiences, such as abuse and neglect. Genetic and biomarker data would provide more direct information about inherited biological traits, while adverse early childhood experiences could potentially influence both parental income and later child outcomes. These are potential source of unobserved confounding, though some of these confounding effects will be indirectly captured through observed parental variables such as mother's age at birth of the child, maternal smoking during pregnancy, highest parental education level, parental disability status, and parental lone parent status.

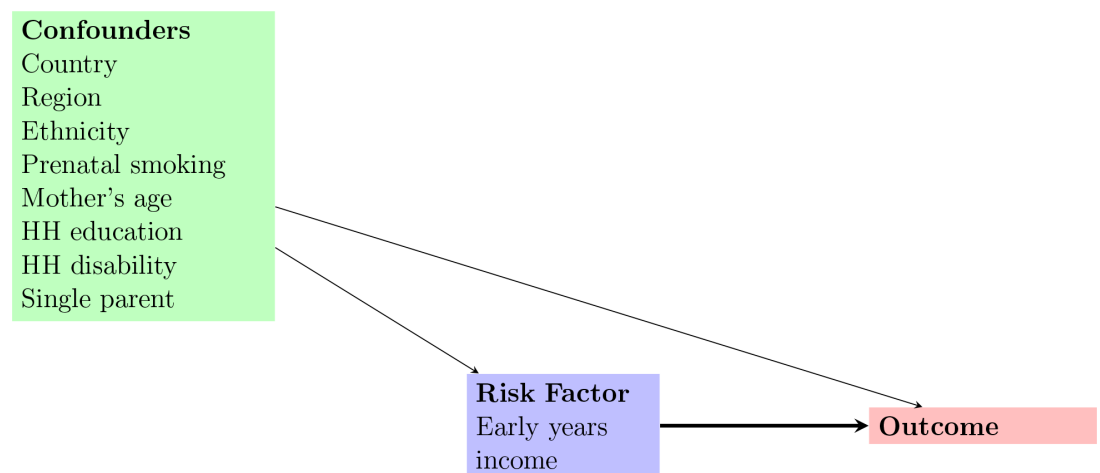
16. Appendix section A.3 provides justification for assuming a causal relationship between early years income and our chosen outcomes

17. The rules along with examples of their application are presented in appendix section A.6

Furthermore, the causal status of many of these unobserved factors is not straightforward. Many inherited influences on later adverse outcomes are realised through cumulative developmental processes involving interactions between biological predispositions, early childhood experiences, and wider material and social environments, rather than through fixed impairments already determined at age 5 (Krieger, 2024; O'Donnell, 2024; Almond et al., 2018; Cunha and Heckman, 2007). Similarly, adverse experiences such as abuse and neglect may partly mediate the effects of low parental income on later outcomes, rather than acting only as pre-existing confounders. Our models include a rich set of observed parental characteristics, family circumstances, and early childhood conditions likely to capture part of these pathways. For comparison, we also estimate unadjusted associations and an “extra conservative” model using an expanded confounder set, including additional variables that may partly lie on the causal pathway. We discuss these issues further in the Discussion.

### 2.3.1 Model used for income as a risk factor

For our illustrative example, we run many different “long-jump” regressions to estimate the effect of early childhood income on many different childhood outcomes at many different ages, using the same basic model specification and control variables in each case. We run the simulation across three versions of our model, our preferred model that follows all the model selection rules, an “extra conservative” model that relaxes some of the rules, and unadjusted correlation. A simplified version of the DAG is presented in **Figure 2**.



**Figure 2.** Simplified directed acyclic diagram (DAG) for the effect of early childhood income

The preferred model controls for everything in the “Confounders” list. We also run an “extra conservative” model which controls for additional variables about which there is room for disagreement because they are partly mediators as well as partly confounders - we call these “partly confounding mediators”. These additional control variables in the extra conservative model are: IMD, maternal MH, HH employment, grandparent’s education, and child’s disability. HH education is highest education in the household, HH disability is an indicator for any parental disability in the household, IMD is the index of multiple deprivation, maternal MH is Maternal malaise measured at 9 months, HH employment is an indicator for labour force participation in the household and the child’s disability is observed at age 3.

The confounders we include in the preferred model are indicators for country (England, Wales, Scotland and Northern Ireland, with England as baseline) at age 9 months, indicators for region within England (with London as baseline) at age 9 months,<sup>18</sup> indicators for ethnicity of the child (with White as baseline),<sup>19</sup> indicator for the mother’s smoking status during pregnancy, age of the mother at birth, indicators for highest education of parents at age 9 months (NVQ levels 1 to 5, with level 5 as baseline), indicator for any disability in the household at age 9 months,<sup>20</sup> and indicator for parent being the single parent at age 9 months (see **Table A11**).

In our extra conservative model we include an additional set of five confounders.

18. North East, North West, Yorkshire and the Humber, East Midlands, West Midlands, East of England, London, South East, and South West

19. White, mixed, Indian, Pakistani and Bangladeshi, Black or Black British, and other ethnic group (inc. Chinese and other Asian)

20. Measured by receipt of any disability benefits by either the main parent or partner.

These include neighbourhood deprivation measured using the Index of Multiple Deprivation (IMD) fifth (with least deprived fifth as baseline) at age 9 months, poor maternal mental health (based on Rutter malaise score) at age 9 months, participation of at least one parent in the labour force at age 9 months, grandparents' education reported at age 17 (NVQ levels 1 to 5, with level 5 as baseline), and disability of the child at age 3.<sup>21</sup>

IMD, maternal mental health, and disability of the child are not included in the main analysis as they may be mediators, i.e. they could all be affected by the parent's permanent income and could affect the child's outcomes.<sup>22</sup> Additionally maternal mental health is also not included because it is measured at 9 months, and post-partum mental health may not be a perfect measure for mental health prior to birth/pregnancy. Parental labour force participation is not included as it is a major determining factor of income and its inclusion may not have a large effect on the child's outcomes. Its inclusion could result in underestimating the effect of income. Finally, grandparent's education is not included in the original model as it is used as a measure of wealth and is unlikely to have a large effect on the child's outcomes except that through the parents' education which we already account for.

The relationship between income and each of the outcomes we simulate is complex. Both risk factor and outcomes are affected by household composition, parental education, parental age, parental physical and mental health, parental wealth, parental socio-economic status, access to resources, child health, and genetics. The richness of data in the MCS about the child and the family allow us to control for many of the potential confounders in these relationships, either directly or indirectly (*Wang et al., 2021*). However, the MCS does not include information on all the potential confounders that may influence both parental income and childhood outcomes, such as genetic traits and early childhood adverse experiences, as discussed further in the Discussion.

### 3. Results

As an illustrative example for this paper we simulate four simple income increasing scenarios, as used in *Villadsen et al. (2023)*.<sup>23</sup> We chose these scenarios as simple thought experiments that illustrate our methodology. These scenarios are based on increasing the income for families grouped by each fifth of the income distribution

- **Scenario 1** - increasing the income for families with children in the poorest fifth to that of the second poorest fifth.
- **Scenario 2** - increasing the income for families with children in the two poorest fifths to that of the middle fifth.
- **Scenario 3** - increasing the income only for families with children in the poorest fifth to that of the richest fifth.
- **Scenario 4** - increasing the income for all families with children to that of the richest fifth.

The weekly mean equivalised household incomes in each fifth of the birth cohort are, £123.47 (£0 to £164) for the poorest, £207.40 (£164 to £255) for the second poorest fifth, £304.54 (£255 to £356) for the middle fifth, £419.29 (£356 to £496) for the next fifth, and £659.05 (£496 to £1,298) for the richest fifth for families with a child born around 2000.<sup>24</sup>

In scenario 1, the equivalised income for the poorest fifth of households increases to £207 a week, an annual average increase of £4,368 a year for these households. On average this cash transfer is about four times the current child benefit amount for single child families, however, families would be eligible for this transfer for only five years instead of between 16 and 20 years and fewer families would be eligible for a cash transfer in scenario 1.

#### 3.1 Adverse Outcomes at Age 17

The simulation estimates that at age 17, 38% of the population have poor GCSEs, 25% experience psychological distress, 26% are obese, 13% are regular smokers, and 10% have poor self reported

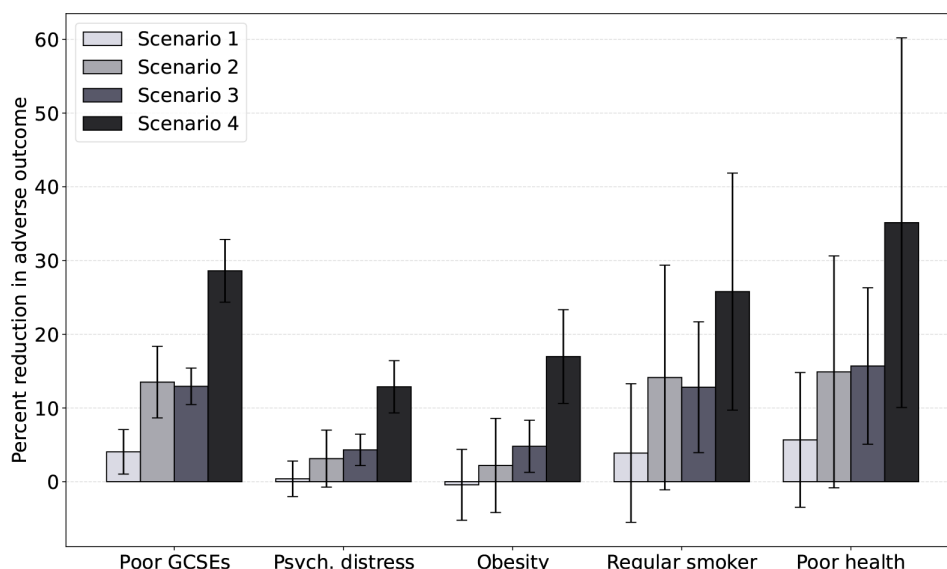
21. This is excluded as a control when estimating the effect of income on disability.

22. Since mediators(M) lie on the causal pathway between the risk factor (RF) and outcome (O), their inclusion in the regression (assuming direction of the effect is the same in all three relationships, RF->O, RF->M, and M->O) will result in smaller estimates of the effect of the risk factor on the outcome.

23. They examined the effects of scenarios 1, 2 and 4 on the five adverse outcomes at age 17.

24. The distribution is presented in Appendix Figure A1.

health.<sup>25</sup> **Figure 3** shows the percentage reduction in these adverse outcome across the entire population associated with each intervention. The error bars show the range in which 95% of the simulated sample effects lie.



**Figure 3.** Percentage reduction in adverse outcomes with each intervention.

Income increase scenarios - Scenario 1 - Poorest to second poorest, Scenario 2 - Poorest two to middle, Scenario 3 - Poorest to richest, Scenario 4 - All to richest. Outcomes are measured at MCS sweep 7 when respondents are around age 17. Poor GCSEs also includes N5 results for Scotland, Psychological distress is identified using the Kessler scale, Obesity is based on UK90 thresholds for sex and age, Regular smoking is more than 6 cigarettes a week.

Scenario 1, increasing the income of the poorest fifth to that of the second poorest fifth results in a 4% (1.6 (0.44 - 2.66) percentage points) reduction in poor GCSEs, 0.4% (0.1 (-0.49 - 0.69) percentage points) reduction in psychological distress, 0.4% (0.1 (-1.10 - 1.31) percentage point) increase in obesity, 3.9% (0.5 (-0.65 - 1.63) percentage point) reduction in regular smoking, and 5.7% (0.6 (-0.3 - 1.44) percentage point) reduction in self reported poor health.

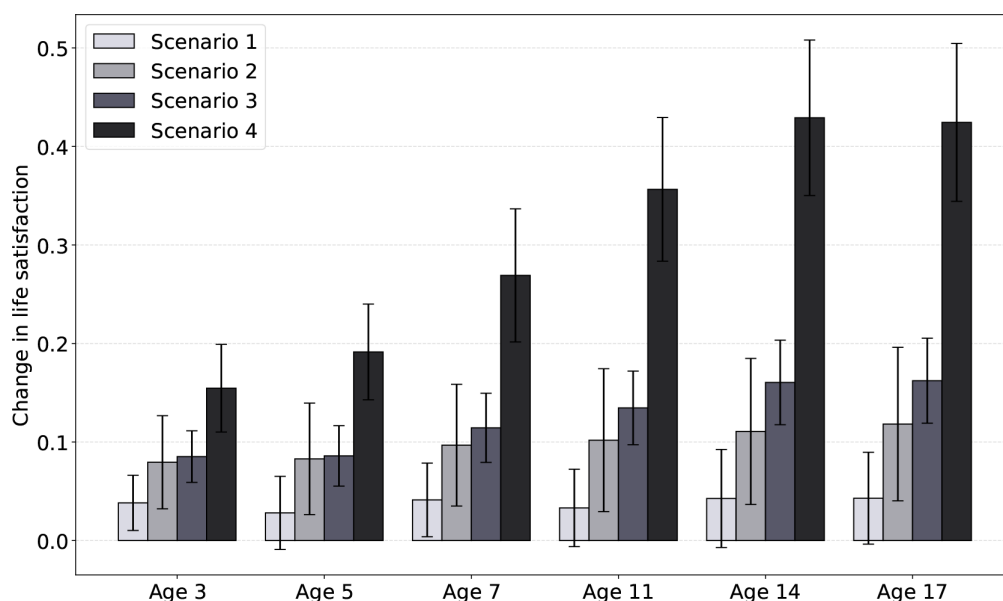
The other scenarios lead to reductions in all five adverse outcomes, including obesity, with larger reductions (percentage) in Poor GCSEs, self reported poor health, and regular smoking than in obesity or psychological distress.

### 3.2 Wellbeing

Wellbeing in our case is measured by parent reported SDQ emotional symptoms scale linearly transformed to life satisfaction on a 0 to 10 scale. The simulated baseline levels of life satisfaction do not fluctuate significantly with age. Life-satisfaction peaks at age 5 with a mean of 7.9 and is lowest at age 17 with a mean of 7.14 (see **Figures A3–A5**).

**Figure 4** shows the average increase in life satisfaction for each scenario, to illustrate the size of the effects on life satisfaction we convert the results into WELLBYs. WELLBYs are defined as a one point increase in life satisfaction on a 10 point scale per child per year. In scenario 1, increasing the income of the poorest fifth to that of the second poorest fifth results in an increase in life satisfaction between 0.028 (-0.009 - 0.065, at age 5) and 0.043 (-0.003 - 0.090, at age 17) in a single year, for an annual average wellbeing increase (between ages 3 and 17) of 0.027 (0.016 - 0.038) or 0.404 (0.238 - 0.570) over the entire 15 year period.

25. The prevalence of these adverse outcomes in the MCS are slightly lower. 37% of the population have poor GCSEs, 23% experience psychological distress, 23% are obese, 10% are regular smokers, and 8% have poor self reported health (see Appendix **Figure A2**)



**Figure 4.** Estimated gain in annual average WELLBYs per child in the general population cohort.

Income increase scenarios - Scenario 1 - Poorest to second poorest, Scenario 2 - Poorest two to middle, Scenario 3 - Poorest to richest, Scenario 4 - All to richest. Life satisfaction is based on a simple linear transformation of parent reported SDQ emotional symptoms score.

The other scenarios lead to greater improvements in wellbeing at every age resulting in a 15 year average wellbeing increase of 1.048 (0.796 - 1.299) in scenario 2, 1.312 (1.170 - 1.455) in scenario 3 and 3.214 (2.906 - 3.522) in scenario 4.

### 3.3 Costs

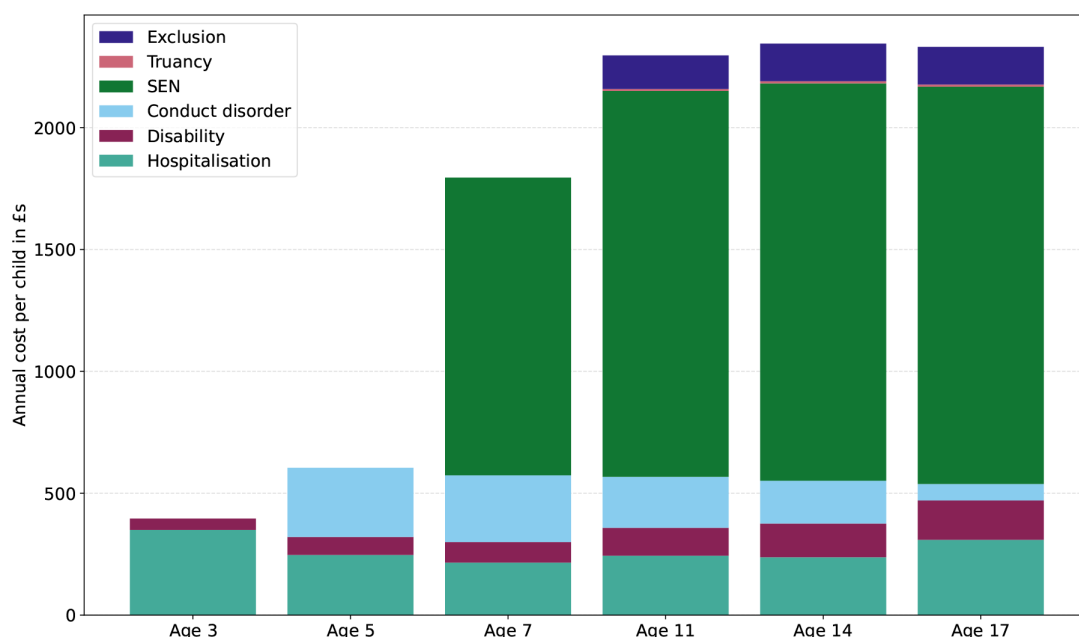
Our public cost estimates capture the costs associated with hospitalisation (ages 3 to 17), disability (ages 3 to 17), conduct disorder (ages 5 to 17), special education needs (ages 7 to 17),<sup>26</sup> persistent truancy (ages 11 to 17),<sup>26</sup> and permanent exclusion (ages 11 to 17).<sup>26</sup>

The annual cost per child at the baseline increases with age as presented in **Figure 5**. The total annual cost per child is £396 at age 3, £604 at age 5, £1,795 at age 7, £2,296 at age 11, £2,345 at age 14, and £2,331 at age 17.<sup>27</sup> This averages to an annual cost per child between the ages of 3 and 17 years of £1,166.<sup>28</sup> This annual cost per child between 3 and 17 can be broken down into £187 for hospitalisation, £71 for disability, £128 for conduct disorder, £727 for special education needs, £3 for persistent truancy, and £50 for permanent exclusion.

26. This outcome is not available at age 17 in the MCS but the value from age 14 is assumed to carry over to age 17.

27. Special education needs, Truancy and Exclusion are not measured at age 17 and are carried over from age 14, the total cost when excluding those three is £538

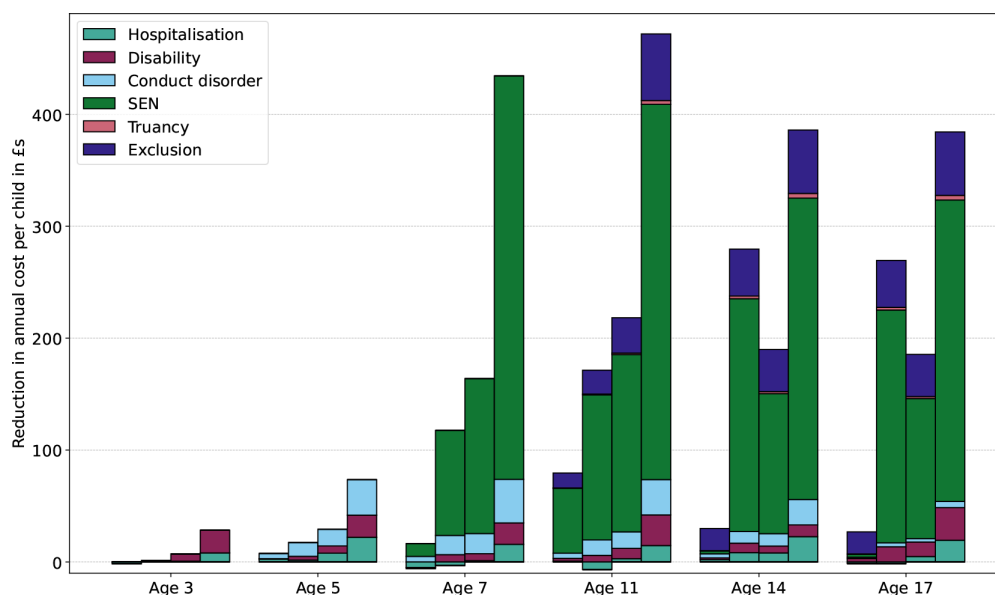
28. Calculated at birth and discounted at 3.5 percent a year.



**Figure 5.** Baseline annual cost per child in each MCS sweep broken down by source.

Truancy, exclusion and special education needs are not measured at age 17 so age 14 values are carried over. Conduct disorder does not include justice costs and the unit cost is lower from age 11.

**Figure 6** shows the cost savings associated with the implementation of each scenario. Note we do not account for the cost of achieving poverty reduction, rather we only show public cost savings. The figure shows four clustered bar graphs at each age, each bar is an estimate for scenarios 1 to 4 in order (from left to right). The area below the x-axis represents cost increases and the area above represents cost savings.



**Figure 6.** Annual cost savings per child for each scenario in each MCS sweep broken down by source.

The bar plots clustered at each age represent scenarios 1, 2, 3, and 4. Income increase scenarios - Scenario 1 - Poorest to second poorest, Scenario 2 - Poorest two to middle, Scenario 3 - Poorest to richest, Scenario 4 - All to richest. Truancy, exclusion and special education needs are not measured at age 17 so age 14 values are carried over. Conduct disorder does not include justice costs and the unit cost is lower from age 11.

Scenario 1 does not result in overall public cost savings (based on our six outcomes) at age 3 due to an increase in hospital costs compared to the baseline. The cohort average annual cost increases by

£1.42 (-13.47 - 16.30) at age 3,<sup>29</sup> and results in savings of £7.60 (-8.15 - 23.35) at age 5, £10.35 (-96.72 - 105.96) at age 7, £79.48 (-59.67 - 158.93) at age 11, £29.87 (-69.90 - 126.32) at age 14, and £25.41 (-76.48 - 123.98) at age 17. This translates to an average annual cost saving between ages 3 and 17 of £18.75 (-15.04 - 41.95),<sup>28</sup> which can be broken down into an increase of £0.41 for hospitalisation, and savings of £0.67 for disability, £2.43 for conduct disorder, £10.19 for special education needs, £0.10 for persistent truancy, and £5.78 for permanent exclusion.

The cost savings in the other scenarios are statistically significant and increase (though there is still an increase in hospitalisation costs in scenario 2) as the money transferred increases. The cohort average annual savings between ages 3 and 17 years<sup>28</sup> are £96.81 (38.72 - 128.54) for scenario 2, £96.37 (67.47 - 126.82) for scenario 3, and £218.87 (115.10 - 322.64) for scenario 4.

While our results are primarily a demonstration of the tool, the increase in hospital utilisation with increase in income, though not significant, may be unexpected. From the literature, there is a plausible mechanism for increased inpatient hospitalisation due to an increase in parental income causing more proactive care seeking behaviour and earlier diagnosis of conditions, resulting in a general increase in non-emergency healthcare utilisation. However, there are very few studies (outside of the

**Table 1.** Simulated effects of scenarios for a UK birth cohort of 700,000 children

| Outcome                                                                                    | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4 |
|--------------------------------------------------------------------------------------------|------------|------------|------------|------------|
| <b>Adverse outcomes (Number of cases prevented by age 17)</b>                              |            |            |            |            |
| Poor GCSEs                                                                                 | 10,909     | 36,366     | 34,831     | 76,999     |
| Psychological distress                                                                     | 697        | 5,576      | 7,685      | 22,916     |
| Obesity                                                                                    | -785       | 4,066      | 8,875      | 31,353     |
| Regular smoker                                                                             | 3,411      | 12,410     | 11,251     | 22,638     |
| Poor health                                                                                | 3,937      | 10,344     | 10,897     | 24,396     |
| <b>Public cost savings (Cost savings between ages 3 and 17, in millions of 2023 £s)</b>    |            |            |            |            |
| Total savings                                                                              | 197        | 1,017      | 1,012      | 2,298      |
| By source                                                                                  |            |            |            |            |
| Any hospitalisation                                                                        | -4         | -3         | 30         | 127        |
| Disability                                                                                 | 7          | 46         | 57         | 155        |
| Conduct disorder                                                                           | 26         | 78         | 84         | 178        |
| Special education needs                                                                    | 107        | 771        | 711        | 1,621      |
| Persistent truancy                                                                         | 1          | 6          | 5          | 13         |
| Permanent exclusion                                                                        | 61         | 118        | 124        | 205        |
| <b>Wellbeing improvement (WELLBYs gained between ages 3 and 17)</b>                        |            |            |            |            |
| WELLBYs                                                                                    | 282,840    | 733,257    | 918,563    | 2,249,815  |
| Value (in millions of 2023 £s)                                                             | 3,677      | 9,532      | 11,941     | 29,248     |
| <b>Scenario cost (Cost to increase income to quintile minimum, in millions of 2023 £s)</b> |            |            |            |            |
| Cost estimate                                                                              | -2,134     | -9,717     | -20,240    | -52,020    |

The number of births in the UK in 2021 was around 700,000. Costs and wellbeing values are discounted at 3.5% per year calculated at birth. Tables with baseline levels and unadjusted figures can be found in the appendix. Income increase scenarios - Scenario 1 - Poorest to second poorest, Scenario 2 - Poorest two to middle, Scenario 3 - Poorest to richest, Scenario 4 - All to richest. Negative values in the table are increases in cases or public costs. Poor GCSEs also includes N5 results for Scotland, Psychological distress is identified using the Kessler scale, Obesity is based on UK90 thresholds for sex and age, Regular smoking is more than 6 cigarettes a week. The program cost is based on minimum the amount that would need to be transferred to each household over five years to move them across fifths of the income distribution.

29. This is due to increases in costs to the public associated this hospitalisation

US and LMICs) looking at the effect of changes to family income on healthcare utilisation. **Reinhold and Jürges (2012)** find that healthcare utilisation among German children has an income gradient. **Coughlan et al. (2022)** find that children living in the most deprived parts of the UK were less likely to use primary and secondary health care but more likely to have emergency visits.

### 3.4 Total effects

**Table 1** summarises the effects of each scenario on a cohort of 700,000 (rough size of 2021 UK birth cohort) children over the full fifteen year period from age 3 to age 17

The first part of the table shows the number of children who avoid each adverse outcome under each simulated scenario as compared to the base-case model.

The next part of the table shows the public cost savings compared to the base-case model in millions of 2023 £s. There is a cost saving of 197 million £s with scenario 1 despite an increase in

**Table 2.** Simulated effects from three models of increasing the income of families with children in the poorest fifth to that of the second poorest fifth for a UK birth cohort of 700,000 children

| Outcome                                                                                   |                           | Preferred Model | Extra Conservative | Unadjusted |
|-------------------------------------------------------------------------------------------|---------------------------|-----------------|--------------------|------------|
| Adverse outcomes ( <i>Number of cases prevented by age 17</i> )                           |                           |                 |                    |            |
|                                                                                           | Poor GCSEs                | 10,909          | 5,965              | 15,068     |
|                                                                                           | Psychological distress    | 697             | -152               | -298       |
|                                                                                           | Obesity                   | -785            | -1,013             | 2,740      |
|                                                                                           | Regular smoking           | 3,411           | 1,527              | 5,837      |
|                                                                                           | Poor Health               | 3,937           | 2,499              | 5,254      |
| Public cost savings ( <i>Cost savings between ages 3 and 17, in millions of 2023 £s</i> ) |                           |                 |                    |            |
|                                                                                           | Total savings             | 197             | -155               | 350        |
| By source                                                                                 | Hospitalisation           | -4              | -17                | 13         |
|                                                                                           | Disability                | 7               | -7                 | 10         |
|                                                                                           | Conduct disorder          | 26              | 15                 | 43         |
|                                                                                           | Special education needs   | 107             | -179               | 212        |
|                                                                                           | Truancy                   | 1               | 1                  | 2          |
|                                                                                           | Exclusion                 | 61              | 32                 | 68         |
| Wellbeing improvement ( <i>WELLBYs gained between ages 3 and 17</i> )                     |                           |                 |                    |            |
|                                                                                           | WELLBYs                   | 282,840         | 180,847            | 435,380    |
|                                                                                           | Value (in millions of £s) | 3,677           | 2,351              | 5,660      |

Number of births in the UK in 2021 was around 700,000. Costs and wellbeing values are discounted at 3.5% per year calculated at birth. Tables with baseline levels and unadjusted figures can be found in the appendix. All results are the estimated impact of income increase shift scenario 1 - Poorest to second poorest. Negative values in the table are increases in cases or public costs. Poor GCSEs also includes N5 results for Scotland, Psychological distress is identified using the Kessler scale, Obesity is based on UK90 thresholds for sex and age, Regular smoking is more than 6 cigarettes a week. The additional control variables in the extra conservative model are: IMD, maternal mental health, household employment, grandparent's education, and child's disability.

hospitalisation costs,<sup>28</sup> driven mostly by savings from reductions in special education needs and school exclusions. The costs savings are higher for the other scenarios.

The third part of the table shows the improvement in wellbeing in terms of WELLBYs, and the value of that increase with WELLBYs being valued at £13,000. For scenario 1 this results in £3,677 million

worth of wellbeing generated over 15 years,<sup>30</sup> for a cohort of 700,000.<sup>28</sup> The amount of wellbeing generated increases with the scale of the money transferred across scenarios.<sup>31</sup>

The final part of the table contains a rough estimate of the cost of the program, based on the minimum amount of money that would have to be transferred to each eligible household to move them to the target fifth of the income distribution. The amount of money that would have to be transferred to all household with newborns over the first five years in the poorest fifth to ensure they have at least as much as the lowest income in the second poorest fifth is £2,134 million.<sup>28, 32</sup>

The public cost savings (up to age 17) from each of the scenarios do not offset their costs. However, for scenario 1 this cost is more than offset by the value gained in terms of wellbeing, before taking into account the public costs savings and savings due to improvement in adverse outcomes and improved outcomes beyond age 17. However, the cost increases with each scenario with Scenario 2's cost just about matching the sum of value generated in wellbeing and the public cost savings. For scenarios 3 and 4 the value of wellbeing generated dwarfs the costs (**Table 1**).

### 3.5 Sensitivity analysis

As a check of our model selection criteria and to get a sense of the effect of controlling for potentially missed confounders we compare our preferred model to an "extra conservative" model and an unadjusted model. **Table 2** summarises the effects of each of the models in scenario 1 on a cohort of 700,000 (rough size of 2021 UK birth cohort) children over the full fifteen year period from age 3 to age 17.

The preferred model shows a reduction in all adverse outcomes except for obesity. In the extra conservative model, there is an increase in psychological distress and a larger increase in obesity than the preferred model, the improvements to the other adverse outcomes are more conservative as may be expected. The unadjusted model shows greater improvements than the preferred model in all adverse outcomes except psychological distress, where there is an increase in the number of cases.

The preferred model suggests with scenario 1 there is a cost savings of £197 million, compared to -£155 with the "extra conservative" model and £350 million, for the unadjusted model. This difference in cost is driven by increases in hospitalisation, disability and special education needs relative to the baseline in the extra conservative model coupled with lower savings on the other outcomes. The savings are higher across all six outcomes and positive across the board in the unadjusted model.<sup>28</sup>

The preferred model shows greater wellbeing gains than the extra conservative model, both of which are much lower than the gains in the unadjusted model. Assuming the cost of implementing scenario 1 is £2,134 million as used above, the gain in value from wellbeing from any of the three models are greater than the cost.

### 3.6 Comparison with other studies

**Villadsen et al. (2023)** look at the effects of scenarios 1, 2 and 4 on adverse outcomes using the MCS. However, they do not include any covariates in their base case estimation of the effects of these income shifting scenarios, and only include two covariates in sensitivity analysis. Consequently, the effects of each of the income shifting scenarios they report, after adjusting for just two covariates, are much larger than our more conservative estimates, roughly varying between 0% (obesity) and 6% (poor health) for scenario 1, between 2% (obesity) and 15% (poor health) for scenario 2, and between 13% (psychological distress) and 35% (poor health) for scenario 4. The general trend and ranking of effect sizes on the adverse outcomes however, is similar to ours.

We use a continuous measure of household income and report findings in terms of effect sizes so that we can compare our estimates with those from a recent systematic review (**Cooper and Stewart,**

30. The Treasury green book (**MacLennan et al., 2021**) suggests the value of a WELLBY is £13,000 with an upper limit of £16,000 and lower limit of £10,000. In which case the wellbeing gain can be valued between £2,828 million and £4,525 million. There is also the much lower supply side valuation of about £2,755 a WELLBY which values the WELLBY gain at £779 million.

31. We also present simulation estimates with SDQ internalising - the sum of emotional and peer problem scores - as the life satisfaction measure instead in **Table A7**. The increase in life satisfaction is about 30% lower for scenario 1 when using this measure.

32. It is likely undertaking a program like this will involve significantly more in administrative costs but we do not calculate them here.

**Table 3.** Effects of increasing income by \$1000 (£676) in Lifesim compared to estimates from **Cooper and Stewart (2021)**

| Cooper and Stewart (2021)                                                 |                    |      |        | LifeSim Childhood      |      |        |
|---------------------------------------------------------------------------|--------------------|------|--------|------------------------|------|--------|
| Paper                                                                     | Outcome            | Age  | Effect | Outcome                | Age  | Effect |
| Blau (1999) ( <i>All families in US</i> )                                 |                    |      |        |                        |      |        |
|                                                                           | PIAT Maths         | 5+   | 0.01   | NFER Progress in maths | 7    | 0.02   |
|                                                                           | PIAT Reading       | 5+   | 0.01   | BAS Word reading       | 7    | 0.02   |
|                                                                           | PPVT               | 3+   | 0.01   | BAS Naming vocabulary  | 3-5  | 0.01   |
| Votruba-Drzal (2006) ( <i>All families in US</i> )                        |                    |      |        |                        |      |        |
|                                                                           | PIAT Maths         | 5+   | 0.02   | NFER Progress in maths | 7    | 0.02   |
|                                                                           | PIAT Reading       | 5+   | 0.02   | BAS Word reading       | 7    | 0.02   |
| Fernald et al. (2008) ( <i>Poor households in Mexico</i> )                |                    |      |        |                        |      |        |
|                                                                           | PPVT               | 4-6  | 0.21   | BAS Naming vocabulary  | 3-5  | 0.03   |
| Milligan and Stabile (2011) ( <i>Low education households in Canada</i> ) |                    |      |        |                        |      |        |
|                                                                           | PIAT Maths         | 6-10 | 0.07   | NFER Progress in maths | 7    | 0.03   |
|                                                                           | Hyperactivity      | 4-10 | 0.07   | SDQ Hyperactivity      | 5-11 | 0.0    |
|                                                                           | Conduct disorder   | 4-10 | 0.10   | SDQ Conduct problems   | 5-11 | 0.03   |
| Dahl and Lochner (2012) ( <i>Poor families in US</i> )                    |                    |      |        |                        |      |        |
|                                                                           | Maths              | 8-14 | 0.21   | NFER Progress in maths | 7    | 0.03   |
|                                                                           | Reading            | 8-14 | 0.21   | BAS Word reading       | 7    | 0.03   |
| Gennetian and Miller (2002) ( <i>Poor families in Minnesota</i> )         |                    |      |        |                        |      |        |
|                                                                           | BPI Internalising  | 5-13 | 0.12   | SDQ Internalising      | 5-14 | 0.03   |
|                                                                           | BPI Externalising  | 5-13 | 0.11   | SDQ Externalising      | 5-14 | 0.02   |
| Dearing et al. (2006) ( <i>Poor households in US</i> )                    |                    |      |        |                        |      |        |
|                                                                           | CBCL Internalising | 2-5  | 0.02   | SDQ Internalising      | 3-5  | 0.03   |
|                                                                           | CBCL Externalising | 2-5  | 0.03   | SDQ Externalising      | 3-5  | 0.01   |
| Zachrisson and Dearing (2015) ( <i>All families in Norway</i> )           |                    |      |        |                        |      |        |
|                                                                           | CBCL Internalising | 2-3  | 0.02   | SDQ Internalising      | 3    | 0.01   |

Effect sizes are based on standard deviation increases in outcome for a \$1000 increase in income. The most similar available test, based on concept measured and age at measurement, in the MCS is used for comparison with those presented in Cooper and Stewart (2021). The age range for the LifeSim estimates are based on the sweep age rather than actual age at time of testing. All MCS measures, except the SDQ scores, are age and ability adjusted measures that are standardised within sample. PIAT - Peabody individual achievement test, NFER - National foundation for educational research, PPVT - Peabody picture vocabulary test, BAS - British ability scales, SDQ - Strength and difficulties questionnaire, BPI - Basic personality inventory, and CBCL - Child behaviour checklist. Ages in MCS correspond to MCS sweeps and not age at observation (age 3 - sweep 2, age 5 - sweep 3 and so on until age 17 - sweep 7).

**2021).** The review compares findings from different studies by reporting the effect of a \$1000 (in year 2000 dollars) increase in income on effect sizes in terms of standard deviation changes for a wide range of outcomes including cognitive development, educational outcomes, child health, and social and behavioural scores. Using LifeSim we estimate the effect of a £676 (Year 2000 equivalent of \$1000) increase in income on some of the human capital outcomes reported by **Cooper and Stewart (2021)** that have an analogue in the MCS. The effect estimates are the change in standard deviation of each outcome with an increase in income of £676. The estimates from **Cooper and Stewart (2021)** and LifeSim are reported in **Table 3** and **Table A10**.

The estimates from LifeSim closely match those from *Blau (1999)*, *Votruba-Drzal (2006)*, *Dearing et al. (2006)*, and *Zachrisson and Dearing (2015)*. However, the estimates from *Fernald et al. (2008)*, *Dahl and Lochner (2012)* and *Gennetian and Miller (2002)* who limit their study to poor households in Mexico, the US and Minnesota, respectively, are much larger than those in LifeSim Childhood. The effect sizes measured based on the equivalent sub-population in the simulation cohort. For example, results of a study focusing on poor households are compared to households in the bottom fifth of the income distribution. However, the estimates used in the simulation are still based on the full population and not specific sub groups. This may partially explain why estimates are more similar for the full population estimates in the systematic review. The estimates from *Cooper and Stewart (2021)* while broadly being similar to those from the MCS are not perfect for the purposes of our comparison as the interventions are different, they are not UK specific and are not all contemporary.

When comparing our MCS-derived estimates to external benchmarks from the existing literature, it is important to note both that we only compare positive outcomes and that published meta-analyses may disproportionately reflect positive outcomes. This can occur due to publication bias against null or negative results, or selective reporting within the reviews themselves, thereby potentially biasing these external benchmark estimates upward.

We estimate a long-term marginal effect of log household income in early childhood on life satisfaction at age 17 of 0.39 on a scale from 0 to 10 (**Table A12**). This implies that a permanent increase of 6.6% in log household income in early childhood - e.g. £1,000 pounds a year to a poor household at the 20th percentile of approximately £15,000 in 2023 - increases life satisfaction on average at age 17 by about 0.026 ( $6.6\% \times 0.39$ ) of one WELLBY. This represents a population average effect of just over one quarter of one percentage point of the maximum life satisfaction score of 10.

Our estimate of 0.39 for the long-term effect of log childhood income on life satisfaction is similar in magnitude to the current best estimate of the long-term effect of log adulthood income on life satisfaction of about 0.38, according to two authoritative reviews of the literature by world leading experts on life satisfaction (*Clark et al., 2019; Frijters and Krekel, 2021*). The estimate of about 0.38 comes from a study by *Lindqvist et al. (2020)* of long-term outcomes for 3,362 large-prize lottery winners randomly selected from a large group of Swedish lottery players. This estimate is more generalisable to the general population than studies of small numbers of large-prize lottery winners randomly selected from a population of committed gamblers. This estimate of about 0.38 is also in line with estimates from analysis of observational data in 113 countries, which found that this estimate was remarkably robust across countries, within countries, and over time (*Stevenson and Wolfers, 2008*). Furthermore, one might expect that increased income during a sensitive period of childhood development would be more beneficial than increased income during adulthood. If anything, therefore, our estimate of 0.39 for the effect of log income in childhood is not only modest but might plausibly be considered an under-estimate.

## 4. Discussion

LifeSim Childhood is a versatile childhood microsimulation model that can be used to estimate the long-term effects of a wide range of childhood risk factors on multiple outcomes up to age 17. In this paper, we describe the methodology used to create LifeSim Childhood, including the model structure, data, estimation strategy and simulation methods. We illustrate its use by estimating the long-term effects of hypothetical increases in early childhood parental income.

### Principal findings

In our illustrative example, we find that increased early childhood income substantially reduces adverse outcomes and improves wellbeing up to age 17. However, estimated public cost savings are relatively small compared with the monetary value of the wellbeing gains. For a cohort of 700,000 newborns, scenario 1 generates estimated public cost savings of £197 million up to age 17, discounted at 3.5%, compared with a gain of 282,840 WELLBYs. Valued at the standard UK Treasury value of £13,000 per WELLBY in 2019 prices, these wellbeing gains are worth £3,677 million. The monetary value of wellbeing gains is therefore about 20 times larger than the estimated public cost savings.

Assuming a hypothetical cost of £2,134 million over five years, scenario 1 results in an estimated net benefit of £1,740 million over 15 years for a single birth cohort. This benefit is almost entirely due to wellbeing gains rather than public cost savings. The cost per WELLBY is £7,545, below both the

Treasury's central value of £13,000 per WELLBY in 2019 prices and its lower bound value of £10,000. However, this should not be interpreted as an evaluation of a directly implementable policy. The scenario is a stylised illustration of the model and may understate the cost of real-world implementation, since it does not include administrative costs or deadweight losses and unrealistically assumes perfect targeting of additional income on families in the poorest fifth. On the other hand, the benefit side is also conservative in important respects, since we only include wellbeing gains for one child in the household, exclude potential effects on parents, siblings and other household members, and do not capture potential public costs savings from reduced numbers of children being taken into care and having other rare but costly adverse experiences that are not well captured in survey data.

The more ambitious income scenarios generate proportionately larger public cost savings and wellbeing gains. In our extra conservative model, with five additional controls variables that do not meet our model selection criteria, we still find substantial estimated total public cost savings and benefits to health, wellbeing and other life outcomes. However, we find slightly larger increases in hospitalisation costs, disability and special educational needs, though none of these estimates are statistically significant. These increases may reflect increased identification of need for these services, but our sample size and statistical power do not allow robust conclusions about the existence or magnitude of such effects.

Finally, we compare LifeSim Childhood estimates with estimates from the literature of the effects of a \$1,000 increase in income on cognitive and socio-emotional scores in the US, Canada, Mexico and Norway. Our estimates are broadly consistent with this literature, particularly for full-population effects. Some studies focusing specifically on poorer groups find larger effects than those estimated using LifeSim Childhood.

## Strengths

LifeSim Childhood has several strengths. First, it allows researchers and policy analysts to estimate effect magnitudes for a range of early childhood risk factors across multiple outcomes up to age 17, including outcomes relevant to health, wellbeing, education and public costs. This is important because many early childhood policies have long-term and cross-sectoral consequences that are difficult to capture using conventional short-term evaluation methods.

Second, the model is based on the Millennium Cohort Study (MCS), a detailed, high-quality and nationally representative longitudinal dataset following more than 15,000 individuals born in the UK in 2000/01 from infancy to age 17. This allows LifeSim Childhood to model developmental pathways across the whole of childhood using repeated measures of family circumstances, child development and later outcomes.

Third, our estimation strategy uses observational data to estimate effect magnitudes for relationships that are already supported by external theory and evidence. The modelling strategy is based on pre-existing scientific understanding of relevant causal pathways, supported by a conservative approach to confounder selection and validation against estimates from a previously published systematic review.

Fourth, LifeSim Childhood uses simulation modelling to synthesise estimates from MCS with unit costs, wellbeing valuations and other external information (**Table A2**). This makes it possible to estimate long-term consequences for multiple outcomes and sectors, while also representing uncertainty. Finally, outputs from LifeSim Childhood can be linked to LifeSim Adulthood to simulate outcomes over the full life course.

## Limitations

The main limitation is that we do not use trial or quasi-experimental data to estimate the effect magnitude for each exposure-outcome pair. Comprehensive and consistent data of this kind, covering the full range of childhood risk factors and outcomes relevant to the UK context, are not available and are unlikely to become available in the foreseeable future. We therefore use observational data from MCS to parameterise our model, and our estimates are limited by the confounders available in that dataset.

LifeSim Childhood may overestimate effect magnitudes if important unobserved confounders are omitted. In practice, adjustment for additional confounding variables usually reduces estimated effects, and the unadjusted model produces larger estimated benefits across all outcomes. However,

when we compare our estimated magnitudes with those from experimental and quasi-experimental studies, we do not find evidence of substantial overestimation. Insofar as omitted-variable bias is broadly similar across risk factors, this concern is also less problematic when LifeSim Childhood is used for relative comparisons of effect magnitude across different childhood risk factors, though this assumption may not always hold.

A specific limitation of MCS is the absence of genetic variables, biomarkers and direct measures of early childhood adverse experiences which are too rare or sensitive to be recorded accurately in survey data, such as death of a parent, bullying, violence, sexual abuse, or being a victim of a crime. Both inherited biological traits and exogenous early childhood adverse experiences could potentially influence both parental income and later childhood outcomes and hence act as unobserved confounders. However, the causal status of many of these potential confounding variables is not straightforward. Many inherited influences on later adverse outcomes are not expressed as fixed impairments already determined at age 5, but are realised through cumulative developmental processes involving interactions between biological predispositions and wider material and social environments. Similarly, many adverse experiences such as parental abuse and neglect may partly mediate the effects of low parental income on later outcomes, rather than acting only as pre-existing confounders.

Our models include a rich set of observed parental and family characteristics that may indirectly capture some of these unobserved pathways, including mother's age at birth of the child, maternal smoking during pregnancy, highest parental education level, parental disability status, and parental lone parent status. For comparison, we also estimate unadjusted associations and an "extra conservative" model using an expanded confounder set which includes additional variables that do not meet our model selection criteria and may lead to under-estimation (e.g. maternal mental health, which may be a mediator that captures part of the total effect of parental income). We return to these issues below when considering the interpretation of LifeSim Childhood estimates.

A second limitation is that our public cost estimates are incomplete. They omit important rare but expensive outcomes, such as being taken into care and contact with the justice system. They also exclude some non-inpatient health care costs, costs associated with disability including social care and healthcare, and costs associated with school exclusion. LifeSim Childhood may therefore under-estimate public cost savings.

Third, LifeSim Childhood focuses on outcomes for a single child, rather than for the entire household, and does not capture spillover effects on parents, siblings and classmates. This is important for household-level risk factors such as parental income, as our illustrative example does not capture the effects of parental income on the wellbeing of siblings or the parents themselves. It is also important for wellbeing analysis more generally, since impact on a child's development and wellbeing may affect the wellbeing of other people.

A fourth limitation is potential cohort bias. We estimate effect magnitudes for Generation Z, born in 2000/01, but apply these estimates to Generation Alpha, born today. There is therefore a risk that relationships between early childhood risk factors and later outcomes have changed across cohorts due to generational changes in family life, public services, education, labour markets, health care, social norms and the wider policy environment.

Fifth, direct data on child-reported life satisfaction on a 0 to 10 scale were not available consistently from ages 3 to 17. We therefore use a proxy measure based on parent-reported SDQ Emotional Symptoms Score, which is measured consistently from ages 3 to 17. Sensitivity analyses using inconsistently measured child-reported subjective wellbeing measures at ages 11, 14 and 17 produce slightly lower WELLBY estimates (see **Table A12**), while analyses using parent-reported SDQ Internalising Score produce slightly higher estimates (see **Table A7**). Our central estimates were broadly consistent with external evidence on adult wellbeing effects (see also **tables A6, A8 and A9**).

Sixth, our probabilistic sensitivity analysis assumed that estimation errors around each causal parameter were all independent - i.e. zero covariance among the errors - since we do not have a methodology for estimating covariance structures among large numbers of separate estimates. In reality, however, covariance is unlikely to be zero: if one parameter is higher than our mean estimate then various other parameters are also likely to be higher. Our simplifying assumption may therefore tend to under-estimate overall uncertainty, insofar as independent errors may tend to cancel out more than correlated errors.

A further limitation is uncertainty about the relationship between exposure duration and long-term effect. For example, to obtain the estimated long-term effect of increased parental income in early childhood, the increase may need to last for one year, five years, ten years or longer. Similar issues arise for other exposures where the relevant duration is uncertain. This is a general challenge for extrapolating long-term intervention effects and is closely related to the problem of effect fadeout. In our illustrative income example, we sidestep this issue by not specifying the duration of change from one income quintile to another. In real-world applications, however, users would need to specify the intervention exposure duration and make assumptions about effect fadeout if the exposure is not permanent.

## Interpretation

These findings should not be interpreted as showing that any specific income transfer would produce exactly the estimated effects, or that public cost savings alone are sufficient to justify early childhood income policies. Rather, the results illustrate how LifeSim Childhood can combine cohort evidence, theory-informed estimates of effect magnitudes, simulation modelling, unit costs and wellbeing valuation to estimate the potential long-term consequences of changes in early childhood risk factors.

The illustrative income example also highlights the importance of including wellbeing outcomes alongside public costs. In this application, the estimated economic case for increasing early childhood income depends primarily on wellbeing gains rather than fiscal savings to public services. This finding is important for policy appraisal, since interventions that improve children's life chances may be undervalued if assessment focuses only on short-term service use or public expenditure.

## Future Work

Future work will address several of these limitations. First, we aim to improve and extend our unit cost estimates, including better modelling of the distribution of costs for each outcome and incorporation of rare but expensive outcomes such as being taken into care and contact with the justice system, using administrative data where possible.

Second, we plan to improve and validate our wellbeing measurement. This will include using other self-reported wellbeing measures available from age 11 onward, refining the mapping between childhood mental health and life satisfaction, and validating our current mapping using other UK datasets.

Third, we aim to assess the extent of cohort bias and update model parameters using the most recent estimates available from external data sources. This is important for applying estimates derived from children born in 2000/01 to children born today.

Fourth, our current models do not capture interactions between multiple risk factors or moderation of effects across population subgroups. Future work will explore the interrelationships between risk factors, beginning with moderation effects. This will allow LifeSim Childhood to provide more nuanced estimates for different population groups and intervention contexts.

## Policy implications

LifeSim Childhood can help researchers and policy makers evaluate a wide range of early childhood interventions and policy scenarios. Users can map the short-term effects of an intervention onto one or more LifeSim Childhood risk factors, describe the relevant population of interest, and estimate long-term consequences for public spending, wellbeing and other life outcomes. Relevant applications could include not only income transfers but also early years services, parenting support, child nutrition programmes, school readiness interventions and child mental health support.

These methods are valuable because policy analysts often find it difficult to quantify the full long-term and cross-sectoral consequences of early childhood policies. LifeSim Childhood provides a transparent framework for estimating these consequences and for comparing the likely long-term impacts of different ways of investing in childhood. In doing so, it can help strengthen the economic case for early intervention while also clarifying whether the main expected benefits arise through public cost savings, improved wellbeing, better educational outcomes, or broader improvements in children's life chances.

## Use of generative AI and AI-assisted technologies

No AI was used in the writing of this paper. AI assistance was used in debugging some of the code for the LifeSim Childhood model.

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## Acknowledgements

We would like to thank all the many people who have contributed towards the development of LifeSim Childhood over many years, through discussions or comments on previous versions of the paper, including the anonymous reviewers, Lucy Andrews, Sarah Bates, Patryk Bronka, Adam Bricknell, Emma Brookes, Eric John Brunner, Sarah Cattán, Mary-Alice Doyle, Sara Evans-Lacko, Rowena Jacobs, Brooke Johnson, Isaac J Parkes, Liina Mansukoski, Emil Margrain, Laurie McClymont, Matteo Richiardi, Matthew Warburton, John Wright, Emir Zaidi, and seminar participants at Health Economics Study Group Meeting Summer 2024, University of Aberdeen, University of Leeds and University of York. We also thank participants of the Millennium Cohort Study for providing valuable data on their lives and experiences.

## Funding

This work was supported by the Medical Research Council Prevention Research Partnership ("ActEarly Programme", Grant # MR/S037527/1) and UK Research and Innovation (Grant # MMR/X002837/1).

## Conflict of Interest

No competing interests reported.

## Data and code availability

The LifeSim Childhood model code is open-source, and distributed freely through GitHub ([https://github.com/LifeSimUK/LifeSim\\_childhood](https://github.com/LifeSimUK/LifeSim_childhood)). This study used secondary data from the Millennium Cohort Study (MCS) sweeps 1-7. MCS data are available to registered researchers from the UK Data Service (SN 2000031) under standard End User Licence (<https://datacatalogue.ukdataservice.ac.uk/series/series/2000031?id=2000031#access-data>). Due to the incorporation of MCS-derived variables, the LifeSim simulated individual-level outputs cannot be made publicly available. However, everything needed to replicate our results is publicly available, including the MCS source data, multiple imputation code, and simulation code.

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## Appendix A

### A Supplementary material

#### A.1 Definitions

1. Risk factors are early childhood targets for policy intervention with long-term consequences to be extrapolated into the future using LifeSim; they may include protective factors with “good” consequences as well as adverse factors with “bad” consequences.<sup>33</sup>
  - a. E.g. when examining the effect of low birth weight on school results, low birth weight is the risk factor.
2. Outcomes are the policy relevant long-term consequences of risk factors,<sup>34</sup> i.e. the measures of the long term effects of changes to risk factors.
  - a. E.g. when examining the effect of low birth weight on school results, school results are the outcome.
3. Confounders are factors that have an independent causal effect on both the risk factor and the outcome. The effect of the confounder on the outcome must be independent i.e. must not entirely go through its effect on the risk factor.<sup>35</sup> Confounders generally occur temporally before both the risk factor and the outcome.
  - a. For example, when examining the effect of low birth weight on school results, parental income is a confounder that influences both the risk of low birth weight and the risk of poor school results. The influence on poor school results is independent, and goes via a number of mechanisms other than its affect on low birth weight.
4. Mediators are on the causal pathway between the risk factor and the outcome. Controlling for mediators will result in biased estimates of the total causal impact of the risk factor, since it will remove indirect effects of the risk factor that are mediated via the effect on the mediating factor.
  - a. E.g. when examining the effect of low birth weight on school results, low birth weight will affect school readiness which in turn will affect school results. School readiness is a mediator in this relationship.
5. Moderators influence the size of the causal effect of the risk factor on the outcome. Moderators generally are temporally before the outcome. Allowing for moderators is not necessary if we are primarily interested in the general population average effect of the risk factor on the outcome, rather than the conditional average effect for a subset of the population with a specific level of the moderating factor.
  - a. E.g. when examining the effect of low birth weight on school results, parental income may influence the impact of low birth weight on school results by influencing access to support, thus mitigating harmful effects for the best off and exacerbating harmful effects for the worst off.

#### A.2 MCS variables

33. They can also be considered as short-term effects from intervention studies that are then harmonised with and plugged into LifeSim.

34. from the point of view of LifeSim, they may be important human capital variables, outcomes that have significant costs to the public, wellbeing, etc.

35. We consider factors that have an effect on the measurement of the risk factor or outcome to also be a confounder, even if they have no effect on the “true” risk factor or outcome.

### A.2.1 Risk Factors

In this paper the risk factor used is early childhood income, the average OECD equivalised income of the household measured around when the child is 9 months, 3 years and 5 years. We use this pooled measure to capture the household's permanent income during early childhood, thus avoiding the risk of bias due to temporary income shocks that would occur if we focused only on income when the child was about 9 months old. The main results specifically look at the effect of increasing income for families to move them up the income distribution.<sup>36</sup> We assume that the timing of change to this "permanent income" starts prior to the point of measurement when the child is approximately 9 months old and continues throughout early childhood to at least the age of five.

In addition to income, LifeSim can be used to evaluate a wide range of other risk factors both in childhood and adolescence such as low birth weight, pre-term birth, disability, school readiness, socio-emotional development, etc.

### A.2.2 Confounders

For the current version of LifeSim with Income as a risk factor, we use a set of eight confounders in the preferred model and an additional five confounders in the "extra conservative" model. We include basic demographic characteristics at age 9 months which we assume to be the same as at birth such as country (England, Wales, Scotland and Northern Ireland) region within England,<sup>37</sup> and ethnicity of the child.<sup>38</sup> Besides this we also include some maternal characteristics, mother's age at birth of the child, and maternal smoking during pregnancy. Finally we include some household characteristics, highest education level of parents (NVQ levels 1 to 5), disability in the household and the child having a single carer.<sup>39</sup>

In the extra conservative model in addition to the variables above we include indicators for the Index of Multiple Deprivation (IMD) fifths (with the least deprived as baseline) at age 9 months, an indicator for poor maternal mental health based on the mother's Rutter Malaise score when the child is 9 months old, an indicator for at least one parent participating in the labour force when the child is 9 months old, indicators for the NVQ education level of the grandparents at measured when the child is 17 years old, and an indicator for the child having a disability at age 3.

### A.3 Relationship between early childhood income and outcomes

In this paper we estimate the effect of income on five outcomes at age 17 (poor GCSEs, Kessler score, obesity, regular smoking, and self reported poor health) and ten outcomes through childhood and adolescence (SDQ emotional symptoms, SDQ conduct problems, SDQ hyperactivity, SDQ peer relationships, hospitalisation, truancy, exclusion, special educational needs statement, disability and cognitive ability). We believe that early childhood income affects each of these outcomes either directly or indirectly through effects to the cognitive ability, socio-emotional/ mental health, or physical health of the child. Two recent systematic reviews of the literature (*Cooper and Stewart, 2021; Page, 2024*), present evidence of the effect of income on cognitive ability. The studies they include show that an increase in income improved cognitive ability as measured by maths, reading and verbal test scores (*Blau, 1999; Dahl and Lochner, 2012; de Gendre et al., 2021; Duncan et al., 2011; Fernald et al., 2008; Milligan and Stabile, 2011; Votruba-Drzal, 2006*), improved cognitive development as measured by memory and visual integration (*Fernald et al., 2008*), improved academic performance in school (*Black et al., 2014; Clark-Kauffman et al., 2003; Duncan et al., 2011; Elstad and Bakken, 2015; Gennetian and Miller, 2002; Votruba-Drzal, 2003*) and increased years of schooling (*Bailey et al., 2024; Bastian and Michelmores, 2018; Votruba-Drzal, 2006*).

36. Income fifths are based on the distribution of household income within the multiply-imputed MCS data with age 3 UK population weights.

37. North East, North West, Yorkshire and the Humber, East Midlands, West Midlands, East of England, London, South East, and South West

38. White, mixed, Indian, Pakistani and Bangladeshi, Black or Black British, and other ethnic group (inc. Chinese and other Asian)

39. Measured by receipt of any disability benefits by either the main parent or partner.

Similarly they present evidence of the effect of income on mental health measured in terms of socio-emotional development. Increasing income during early childhood improves the socio-emotional health of children (*Blau, 1999; Dearing et al., 2006; Gennetian and Miller, 2002; Hamad and Rehkopf, 2015; Votruba-Drzal, 2006; Zachrisson and Dearing, 2015*), improved engagement, and reduced inattention, anxiety and aggression (*Milligan and Stabile, 2011*).

There are fewer studies looking into the causal effect of early years income on health outcomes after birth. Despite a large number associational and descriptive research on the subject, the causal literature is still relatively sparse. Some of the literature looking at the specific physical health outcomes of interest finds no statistically significant effect during childhood but finds an effect in adulthood (*Currie, 2009; Kuehnle, 2014*). *de Gendre et al. (2021)* and *Ko et al. (2020)* find that early years income results in lower likelihood of hospitalisations, and *Doyle et al. (2024)* found a reduction in income lead to increases in hospitalisation. *Baughman and Duchovny (2016)* and *East (2020)* find that early years income increases leads to reduced likelihood of poor/fair health and increased likelihood of excellent health. *Aizer et al. (2022)* show that an increase in income leads to increases life expectancy, increases BMI (but not obesity) reduces probability of being underweight. *East (2020); Ko et al. (2020)* also find early years income leads to lower rates of developmental delay and reduced incidences of acute and chronic conditions up to age 3. Since we define disability as a long term illness that affects day to day activity we believe a reduction in chronic conditions will translate to a reduction in disability among the children as well. The most studied mechanism through which income affects physical health is nutrition, particularly for low income households, *Milligan and Stabile (2011)* provide evidence for this, they show a reduction in reports of children going hungry following an increase in income during early childhood.

Another indirect mechanism we also hope to capture is the indirect effects on the outcomes through effects on the parents, either by increasing their income beyond the transfers (*Barr et al., 2022*), improved parental mental health / reduction in stress (*Jones et al., 2019*), changes in parental behaviour (*Bullinger et al., 2023; Jones et al., 2019; Votruba-Drzal, 2006*).

Our estimates may also be affected by reverse causation, whereby serious developmental difficulties emerging in early childhood reduce parental income at ages 9 months, 3 and 5, rather than parental income affecting child development. For example, caring for a disabled young child might affect parents' mental health, labour market participation, earnings or employment stability more than caring for a non-disabled child. This issue of reverse causation is potentially relevant to three early childhood outcomes measured at ages 3 and 5: disability, hospitalisation and emotional symptoms (our proxy for wellbeing). As a simple check on the potential magnitude of this bias, we repeat our simulation of these three outcomes using only parental income at 9 months as the risk factor and compare this with our base case simulation using five-year average income as the risk factor (see *Table A1*). The assumption underpinning our simple check is that population average parental income loss due to reverse causation will plausibly occur gradually over time, rather than taking immediate and full effect in the first nine months when (i) paid parental leave protections are in place for most parents, (ii) differential caring burdens may not yet have taken their full cumulative toll on many parents, and (iii) many early childhood developmental problems may not yet have manifested by 9 months. If reverse causation bias is materially important, the 9-month income measure should be less contaminated by it than the five-year average, since any channel through which emerging childhood developmental problems affect parental income, whether via employment changes or parental mental health, has had less time to operate by 9 months. The 9-month estimates should therefore show weaker associations with child outcomes at ages 3 and 5 than the five-year estimates if reverse causation is driving our results. Appendix table A13 presents estimates of the effects of both parental income measures on disability, hospitalisation and emotional symptoms at ages 3 and 5. We find that the estimated effects of parental income at 9 months are either higher than or the

same as the estimated effects of five-year average parental income, though no difference is statistically significant. This suggests that reverse causation is unlikely to materially affect our estimates in this instance.

#### A.4 Measurement of wellbeing

The current model uses parent-reported SDQ Emotional Symptoms as the primary measure of wellbeing as parent reported SDQ scores are available consistently across all current sweeps. Alternative wellbeing measures are available at ages 11, 14 (Wellbeing grid) and 17 (SWEMWBS) which we hope to use in the future at those ages and help map the SDQ scores to a better wellbeing measure for us to use at earlier ages.

An alternative that is available at all ages is parent-reported SDQ Internalising, the sum of SDQ emotional symptoms and Peer problems scales. We choose to use SDQ emotional symptoms instead of SDQ internalising for a few reasons. SDQ scores used in our model are parent-reported and therefore may not reflect the actual levels of each attribute but rather the level perceived by the parent. The relationship between the actual levels and parent-reported levels for peer problems may be less correlated than that for emotional symptoms for school-aged children. While we do not have a wellbeing measure, SDQ emotional symptoms scale is predictive of a number of outcomes that are closely related to wellbeing (such as mental disorders, school performance.) whereas the concepts associated with the peer problems scale are more tenuously associated with wellbeing.

A version of Table 1 with wellbeing measured by SDQ internalising instead of just emotional symptoms is presented in appendix table A7.

#### A.5 Multiple Imputation

Of the cohort of 19,519 children in the MCS only 10,757 children provided any data at age 17, this is without accounting for incomplete responses to specific questions even in years they did participate. To deal with this attrition and missingness in the data we use multiple imputation. We believe this helps increase the statistical power and reduce bias in our estimation of effects.

We used chained equations to impute missing data back to the second sweep (age 3) of the MCS. We use the second sweep instead of the first as more than a thousand eligible households were added to the cohort in that sweep that were missed in the first sweep. We follow *Villadsen et al. (2023)* and include a number of the same auxiliary variables that are predictive of our outcomes at age 17 and earlier in the imputation to improve the accuracy of our estimates. These variables are from various sweeps and included child physical health, parental substance use, maternal mental health, and other household characteristics. We generate 30 datasets using multiple imputation. Post-imputation, the final sample used in our analyses consists of 15,380 cohort members.

#### A.6 Model selection rules

To build our models we follow a set of rules, described below with examples of their application, so our models are credible, conservative, parsimonious and focused on population averages. The rules below are used in the current version of LifeSim Childhood, but may change in the future as we develop LifeSim further. We will follow the DAG reporting recommendations of *Tennant et al. (2021)* in future work and include detailed documentation for each DAG produced along with the code.

1. Credible: Confounder selection must be based on strong theoretical framework, scientific evidence and expert opinion. There must be a credible story that justifies why the variable is a well-founded confounder.
2. Conservative: We include confounders that are also partly mediators, even though this will remove part of the total causal effect. Similarly, we also include confounders that are

also moderators.<sup>40</sup> That makes our approach conservative. However, we do not wish to be over-conservative by over-adjusting the total causal effect downwards by controlling for too many mediating effects. So we do not control for “partly confounding mediators” that we consider to be primarily mediators rather than confounders - i.e. we make a scientific judgement that the mediating impact is more important than the confounding impact. This requires a scientific judgement drawing on prior knowledge about the temporal sequence of events and, if necessary, the expected relative strength of the causal impacts and correlations. If there is doubt about whether a variable is “primarily” a confounder or a mediator then we do not include it in the base model but do in the “extra conservative” model as a robustness check.

- E.g. 1 a “partly confounding mediator”: when examining the effect of having a teenage mother on school results at age 17 (GCSEs), it is unclear if maternal mental health (MMH) at age 9 months primarily a confounder or mediator. We judge it is primarily a mediator because we expect variation in MMH age 9 months to be strongly driven by having a baby, and the relevant timing of MMH for causing teen pregnancy is at least 2 years before that at conception, so even though MMH at 9 month may be correlated with MMH 2 years before, the confounding impact is expected to be less strong than the mediating impact.
  - E.g. 2 a “partly mediating confounder”. when examining the effect of having a teenage mother on school results at age 17 (GCSEs), we do control for smoking in pregnancy because this is a proxy for risky behaviour which may confound the teen pregnancy effect. This is debatable, however, insofar as teen pregnancy itself might be conceptualised as related to risky behaviour which partly relates to some of the social causal pathways from teen pregnancy to child outcomes. We therefore do not include this in our preferred model but do include this as a confounder in our “extra conservative” model. (A similar example is whether to control for IMD, which turns on whether pregnancy is more likely to cause the parents to move out of the grandparents house to a deprived neighbourhood, or whether living in a deprived neighbourhood is more likely to cause teenage pregnancy).
  - E.g. 3 a “confounding moderator”. When examining the effect of disability at age 3 on school results at age 17 (GCSEs), income between ages 0 and 5 is included as a confounder even though we expect it to have a moderating effect. This moderating effect is not explicitly modelled because we are not including interaction terms. We just want the population average effect.
3. Parsimonious: We prefer a simple model unless additional complexity is clearly justified and makes an important difference. As well as the general scientific virtue of keeping things as simple as possible but not simpler, parsimony in relation to confounder selection has the further advantage of reducing the variance of the effect estimate: adding further possible confounding variables that are not distinctive from the preferred set of confounding variables will tend to increase the variance of the effect estimate without improving its accuracy.
- a. Use a simple functional form with fewer parameters to estimate and interpret where possible.
  - b. Exclude confounders that are not distinctive i.e. they are likely measuring the same thing as other confounders.
    - i. A confounder is considered distinct if the causal pathway to either the risk factor or the outcome is distinct from other confounders. The causal pathway to both the risk factor and the outcome do not have to be distinct.
    - ii. When multiple options are available choose the measure most relevant for the causal pathway.

40. This also includes moderators that can be considered a special type of “collider” that involves selection into the risk factor but does not involve “collider bias”.

- iii. E.g. when examining the effect of low birth weight on school results at age 17 (GCSEs), being a single parent could be a potential confounder as it could capture some economic characteristics of the household (it likely will not capture anything else, such as time with child because of exposure timing) that may affect birth weight and later education outcomes. However, we include household income and IMD as confounders and that should capture everything that single parent captures in this instance. This will not be the case if the exposure is disability at age 5.
  - c. Exclude prior measurements of outcome and/or risk factor. We are not interested in modelling the effect of changes or the trajectory of outcomes by each sweep, we are interested only in the full direct effect.
    - i. E.g. when examining the effect of cognitive ability at age 7 on wellbeing at age 17, we do not include measures of cognitive ability or wellbeing as confounders. We are currently interested in modelling the direct effect of a specific cognitive ability level on wellbeing and not on how changes in cognitive ability affects wellbeing or cognitive ability affects wellbeing trajectory.
    - ii. Note: An alternative approach to causal inference would be to include a prior measure of the risk factor as a control variable - for example, estimating the effect of income at age 5 controlling for income at age 3. However, we believe this would be an over-adjustment that would under-estimate the causal effect of the exposure level at T on the outcome level at T+1. Including a prior measure would estimate the effect of the exposure trajectory (first difference) rather than the effect of the exposure level which is our parameter of interest.
  - d. Do not include both linear and non-linear forms of the same confounder. But the linear or non-linear measures can be used individually as appropriate for the model.
    - i. E.g. when examining the effect of disability at age 3 on school results at age 17 (GCSEs), income between ages 0 and 5 is included as a confounder. Poverty between ages 0 to 5 is not currently included as a confounder as income and poverty are seen to capture the same causal pathway, even though those below the poverty line may have different effects for those above.
  - e. For an outcome risk factor pair, the model used is specific to risk factor timing only.
    - i.e. for a particular risk factor and outcome pair the same model should be used for outcomes at all ages as long as risk factor timing is constant. However if the risk factor timing changes the model should also change.
    - i. E.g. when examining the effect of cognitive ability at age 7 on wellbeing, we use the same model to simulate wellbeing from ages 11 upto 17. However, if we want to examine the effect of cognitive ability at age 5 on wellbeing, we use a different model to simulate the effects on wellbeing from ages 7 to 17.
- 4. Focused on population averages Do not include interaction terms, since our primary interest lies in the unconditional population average effect rather than the moderating effects for sub-populations with specific levels of the moderating variable.
  - a. E.g. when examining the effect of disability at age 3 on school results at age 17 (GCSEs), we do not include interaction between age 3 disability and parental income age 0-5, since we currently just want the population level effect of age 3 disability and not how this effect is modified by parental income.

## A.7 Advisory group

The advisory group for the two grants from the Medical Research Council Prevention Research Partnership ("ActEarly Programme", Grant # MR/S037527/1) and UK Research and Innovation (Grant # MMR/X002837/1) include those below. We are extremely grateful to all

members of our advisory group for their support, though absolve them of responsibility for our methodological choices and errors.

**Anna Freud:** Jessica Deighton

**Bradford Institute for Health Research:** David Ryan, John Wright

**Department for Education:** Catherine Newsome

**Department of Health and Social Care:** Lucy Andrews

**Department for Work & Pensions:** Mike Daly

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**University of Leeds:** Mark Mon-Williams

**University of Liverpool:** David Taylor-Robinson

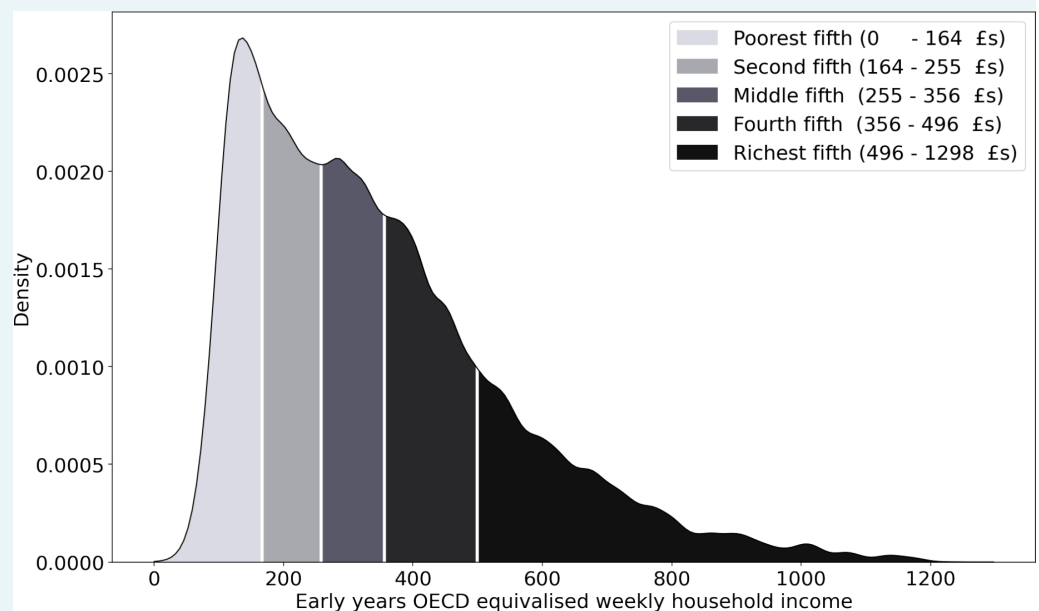
**University of Oxford:** Leon Feinstein

**University of Sheffield:** Sarah Bates, Alan Brennan

**University of York:** Karen Bloor, Jonathan Bradshaw, Richard Cookson, Tim Doran, Ruth Helstrip, Rowena Jacobs, Ruth Patrick, Kate Pickett, Ieva Skarda, Shrathinth Venkatesh

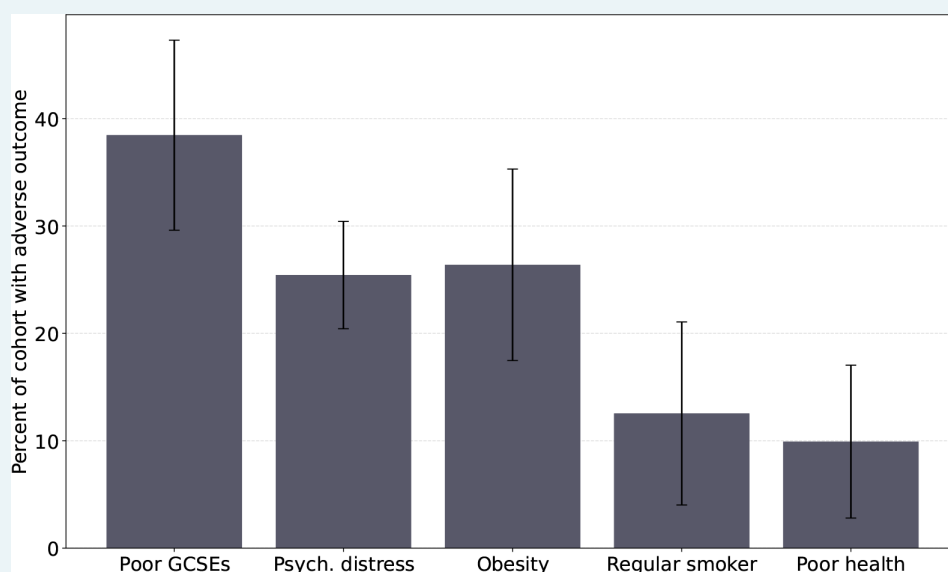
**Young Person Representative:** Brooke Johnson

**Youth Futures Foundation:** Andrea Barry



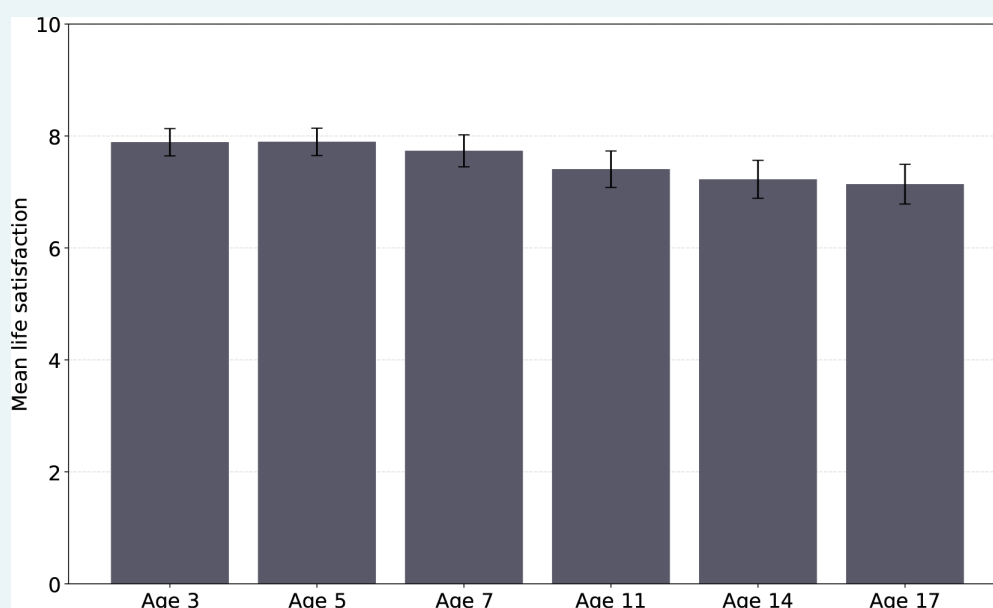
**Figure A1.** Density plot of early years OECD equivalised weekly household income in the MCS split into quintiles.

The means for each quintile or fifth of the distribution are £123.47, £207.40, £304.54, £419.29 and £659.05 respectively.



**Figure A2.** Baseline prevalence of adverse outcomes at age 17.

Outcomes are measured at MCS sweep 7 when respondents are around age 17. In Scotland, Poor GCSEs refers to N5 exam results, psychological distress is identified using the Kessler scale, obesity is based on UK90 thresholds for sex and age, regular smoking is more than 6 cigarettes a week.



**Figure A3.** Baseline life satisfaction proxied by SDQ emotional symptoms score in each MCS sweep.

Life satisfaction is based on a simple linear transformation of parent reported SDQ emotional symptoms scale.

**Table A1.** Summary of unit costs currently used in LifeSim Childhood

| Simulated Outcomes | Ages Available in MCS | Public Cost Sector |        |           |                  |                              | Annual Unit Cost(2023 £) |
|--------------------|-----------------------|--------------------|--------|-----------|------------------|------------------------------|--------------------------|
|                    |                       | NHS                | Social | Education | Criminal Justice | Outcome Costed               |                          |
| Hospital Admission | 3, 5, 7, 11, 14, 17   | X                  |        |           |                  | Inpatient visit for children | 1,587                    |
| Disability         | 3, 5, 7, 11, 14, 17   | X                  |        |           |                  | Per child cost to NHS        | 1,008                    |

| Public Cost Sector      |                     |   |   |   |                                                             |
|-------------------------|---------------------|---|---|---|-------------------------------------------------------------|
| Conduct Disorder        | 3, 5, 7, 11, 14, 17 | X | X | X | Conduct disorder 3,092 (5 to 10) 1,963 (11 to 16) 236 (17+) |
| Special Education Needs | 7, 11, 14           |   |   | X | EHC Plan 25,500                                             |
| Any Truancy             | 11, 14              |   |   | X | Persistent Truancy 943                                      |
| Any Exclusion           | 11, 14              |   |   | X | Alternate Provision 21,848                                  |

Ages in MCS correspond to MCS sweeps and not age at observation (age 3 - sweep 2, age 5 - sweep 3 and so on until age 17 - sweep 7).

**Table A2.** Sources of unit costs currently used by LifeSim childhood

| Simulated Outcome          | MCS Source       | Cost Outcome          | Ages costed | Cost year | Cost source                                                                                                                                          |
|----------------------------|------------------|-----------------------|-------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hospital Admission         | Parent admission | Hospital              | 0-18        | 2020      | Average cost of a single inpatient hospitalisation for a child aged between 0 and 17 in the UK in 2003/2004 at 2019/2020 prices. (Dale et al., 2024) |
| Disability                 | Parent           | Healthcare            |             | 2016      | average annual cost of healthcare for a child in the UK. (Kelly et al., 2018)                                                                        |
| Conduct disorder           | Parent           | Conduct disorder      | 5 to 18     | 2008      | Costs from 4 papers published between 2000 and 2007. (Bonin et al., 2011)                                                                            |
| Special Educational Needs  | Teacher          | EHC Plan              | 5 to 18     | 2023      | Average incremental cost to DfE of student with EHC plan in 2023 (Acton et al., 2024)                                                                |
| Persistent truancy         | CM               | Persistent Truancy    | 11 to 12    | 2005      | Cost education welfare services in England and number of persistent truants in 2005. (Brookes et al., 2007)*                                         |
| Permanent School Exclusion | Parent           | Alternative Provision | 5 to 18     | 2018      | Survey of 101 LAs by ISOS for the Department for Education in FY 2017-2018. (Bryant et al., 2018)                                                    |

\* also source for costs in the Greater Manchester unit cost database.

**Table A3.** Simulation population descriptive statistics for confounders and exposure

|           | Characteristic           | Mean/Proportion | Standard Error |
|-----------|--------------------------|-----------------|----------------|
| Sex       | Male                     | 0.503           |                |
|           | White                    | 0.860           |                |
|           | Mixed                    | 0.032           |                |
|           | Indian                   | 0.018           |                |
|           | Pakistani or Bangladeshi | 0.044           |                |
|           | Black                    | 0.027           |                |
| Ethnicity | Other                    | 0.013           |                |
|           | England                  | 0.826           |                |
|           | Wales                    | 0.047           |                |
|           | Scotland                 | 0.091           |                |
| Country   | Northern Ireland         | 0.037           |                |

|                             | Characteristic             | Mean/Proportion | Standard Error |
|-----------------------------|----------------------------|-----------------|----------------|
| Region                      | North East                 | 0.037           |                |
|                             | North West                 | 0.103           |                |
|                             | Yorkshire and the Humber   | 0.089           |                |
|                             | East Midlands              | 0.071           |                |
|                             | West Midlands              | 0.081           |                |
|                             | East of England            | 0.088           |                |
|                             | London                     | 0.130           |                |
|                             | South East                 | 0.148           |                |
|                             | South West                 | 0.078           |                |
| Household Highest Education | No qualifications          | 0.119           |                |
|                             | NVQ level 1                | 0.081           |                |
|                             | NVQ level 2                | 0.340           |                |
|                             | NVQ level 3                | 0.098           |                |
|                             | NVQ level 4                | 0.293           |                |
|                             | NVQ level 5                | 0.037           |                |
| Household Characteristics   | Disability in household    | 0.029           |                |
|                             | Single parent              | 0.146           |                |
|                             | Age of mother (at birth)*  | 28.723          | 0.041          |
|                             | Annual early years income* | 18226.072       | 74.836         |

\*mean and standard error, all other numbers are proportion of population in category.

**Table A4.** Simulated baseline outcome means

| Outcome                 | Age 3           | Age 5           | Age 7           | Age 11          | Age 14          | Age 17          |
|-------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Poor GCSEs              |                 |                 |                 |                 |                 | 0.38<br>(0.045) |
| Psychological distress  |                 |                 |                 |                 |                 | 0.25<br>(0.026) |
| Obesity                 |                 |                 |                 |                 |                 | 0.26<br>(0.046) |
| Regular smoker          |                 |                 |                 |                 |                 | 0.13<br>(0.044) |
| Poor health             |                 |                 |                 |                 |                 | 0.10<br>(0.036) |
| Life satisfaction       | 7.89<br>(0.125) | 7.90<br>(0.125) | 7.73<br>(0.146) | 7.41<br>(0.166) | 7.23<br>(0.173) | 7.14<br>(0.181) |
| Any hospitalisation     | 0.22<br>(0.034) | 0.16<br>(0.033) | 0.14<br>(0.032) | 0.15<br>(0.031) | 0.15<br>(0.041) | 0.19<br>(0.043) |
| Disability              | 0.05<br>(0.022) | 0.07<br>(0.024) | 0.08<br>(0.026) | 0.11<br>(0.033) | 0.14<br>(0.033) | 0.16<br>(0.046) |
| Conduct disorder        |                 | 0.09<br>(0.005) | 0.09<br>(0.005) | 0.09<br>(0.006) | 0.09<br>(0.007) | 0.08<br>(0.006) |
| Special education needs |                 |                 | 0.05<br>(0.030) | 0.06<br>(0.028) | 0.06<br>(0.030) |                 |
| Any truancy             |                 |                 |                 | 0.04<br>(0.031) | 0.05<br>(0.025) |                 |

| Outcome       | Age 3 | Age 5 | Age 7 | Age 11          | Age 14          | Age 17 |
|---------------|-------|-------|-------|-----------------|-----------------|--------|
| Any exclusion |       |       |       | 0.06<br>(0.066) | 0.09<br>(0.043) |        |

Terms in brackets are the standard deviation of the distribution of means across simulations. Ages correspond to MCS sweeps and not age at observation (age 3 - sweep 2, age 5 - sweep 3 and so on until age 17 - sweep 7).

**Table A5.** Simulated baseline cost means by source

| Outcome                 | Age 3             | Age 5             | Age 7               | Age 11              | Age 14              | Age 17              |
|-------------------------|-------------------|-------------------|---------------------|---------------------|---------------------|---------------------|
| Any hospitalisation     | 349.49<br>(54.33) | 246.68<br>(51.84) | 215.45<br>(51.02)   | 243.71<br>(49.26)   | 237.38<br>(65.29)   | 308.90<br>(67.54)   |
| Disability              | 46.67<br>(21.74)  | 73.89<br>(24.25)  | 84.12<br>(26.00)    | 114.43<br>(33.64)   | 138.30<br>(33.13)   | 161.57<br>(46.54)   |
| Conduct disorder        |                   | 283.86<br>(16.04) | 273.90<br>(16.19)   | 209.21<br>(13.89)   | 175.65<br>(13.49)   | 67.39<br>(4.48)     |
| Special education needs |                   |                   | 1221.48<br>(758.85) | 1583.28<br>(697.62) | 1630.01<br>(770.40) | 1630.01<br>(770.40) |
| Permanent exclusion     |                   |                   |                     | 137.77<br>(148.40)  | 154.34<br>(73.18)   | 154.34<br>(73.18)   |
| Persistent truancy      |                   |                   |                     | 7.79<br>(5.37)      | 8.91<br>(4.38)      | 8.91<br>(4.38)      |

Costs for special education needs, permanent exclusion, and persistent truancy are carried over from age 14 to age 17 as those outcomes are not measured at age 17. Terms in brackets are the standard deviation of the distribution of means across simulations. Ages correspond to MCS sweeps and not age at observation (age 3 - sweep 2, age 5 - sweep 3 and so on until age 17 - sweep 7).

**Table A6.** Simulated undiscounted effects of scenarios for a UK birth cohort of 700,000 children

| Outcome                                                                                    | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4 |
|--------------------------------------------------------------------------------------------|------------|------------|------------|------------|
| <b>Adverse outcomes (Number of cases prevented by age 17)</b>                              |            |            |            |            |
| Poor GCSEs                                                                                 | 10,909     | 36,366     | 34,831     | 76,999     |
| Psychological distress                                                                     | 697        | 5,576      | 7,685      | 22,916     |
| Obesity                                                                                    | -785       | 4,066      | 8,875      | 31,353     |
| Regular smoker                                                                             | 3,411      | 12,410     | 11,251     | 22,638     |
| Poor health                                                                                | 3,937      | 10,344     | 10,897     | 24,396     |
| <b>Public cost savings (Cost savings between ages 3 and 17, in millions of 2023 £s)</b>    |            |            |            |            |
| Total savings                                                                              | 296        | 1,574      | 1,512      | 3,394      |
| By source                                                                                  |            |            |            |            |
| Any hospitalisation                                                                        | -5         | -3         | 44         | 181        |
| Disability                                                                                 | 11         | 69         | 82         | 217        |
| Conduct disorder                                                                           | 36         | 109        | 117        | 248        |
| Special education needs                                                                    | 155        | 1,197      | 1,062      | 2,404      |
| Persistent truancy                                                                         | 2          | 10         | 9          | 21         |
| Permanent exclusion                                                                        | 97         | 192        | 198        | 324        |
| <b>Wellbeing improvement (WELLBYs gained between ages 3 and 17)</b>                        |            |            |            |            |
| WELLBYs                                                                                    | 398,629    | 1,042,533  | 1,326,571  | 3,293,455  |
| Value (in millions of 2023 £s)                                                             | 5,182      | 13,553     | 17,245     | 42,815     |
| <b>Scenario cost (Cost to increase income to quintile minimum, in millions of 2023 £s)</b> |            |            |            |            |
| Cost estimate                                                                              | -2,364     | -10,760    | -22,414    | -57,607    |

| Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|------------|------------|
| <b>Adverse outcomes (Number of cases prevented by age 17)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |            |            |            |
| Number of births in the UK in 2021 was around 700,000. Tables with baseline levels and unadjusted figures can be found in the online appendix. Income increase scenarios - Scenario 1 - Poorest to second poorest, Scenario 2 - Poorest two to middle, Scenario 3 - Poorest to richest, Scenario 4 - All to richest Negative values in the table are increases in cases or public costs. Poor GCSEs also includes N5 results for Scotland, Psychological distress is identified using the Kessler scale, Obesity is based on UK90 thresholds for sex and age, Regular smoking is more than 6 cigarettes a week. |            |            |            |            |
| <b>Table A7. Simulated effects of scenarios for a UK birth cohort of 700,000 children with SDQ internalising as the wellbeing measure</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |            |            |            |
| Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4 |
| <b>Adverse outcomes (Number of cases prevented by age 17)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |            |            |            |
| Poor GCSEs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 10,909     | 36,366     | 34,831     | 76,999     |
| Psychological distress                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 697        | 5,576      | 7,685      | 22,916     |
| Obesity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -785       | 4,066      | 8,875      | 31,353     |
| Regular smoker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3,411      | 12,410     | 11,251     | 22,638     |
| Poor health                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3,937      | 10,344     | 10,897     | 24,396     |
| Public cost savings (Cost savings between ages 3 and 17, in millions of 2023 £s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |            |            |            |
| Total savings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 197        | 1,017      | 1,012      | 2,298      |
| By source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |            |            |            |
| Any hospitalisation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -4         | -3         | 30         | 127        |
| Disability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 7          | 46         | 57         | 155        |
| Conduct disorder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 26         | 78         | 84         | 178        |
| Special education needs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 107        | 771        | 711        | 1,621      |
| Persistent truancy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1          | 6          | 5          | 13         |
| Permanent exclusion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 61         | 118        | 124        | 205        |
| Wellbeing improvement (WELLBYs gained between ages 3 and 17)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |            |            |            |
| WELLBYs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 197,879    | 629,453    | 678,858    | 1,656,902  |
| Value (in millions of 2023 £s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2,572      | 8,183      | 8,825      | 21,540     |
| Scenario cost (Cost to increase income to quintile minimum, in millions of 2023 £s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |            |            |            |
| Cost estimate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -2,134     | -9,717     | -20,240    | -52,020    |

Number of births in the UK in 2021 was around 700,000. Costs and wellbeing values are discounted at 3.5% per year calculated at birth. Tables with baseline levels and unadjusted figures can be found in the online appendix. Income increase scenarios - Scenario 1 - Poorest to second poorest, Scenario 2 - Poorest two to middle, Scenario 3 - Poorest to richest, Scenario 4 - All to richest Negative values in the table are increases in cases or public costs. Poor GCSEs also includes N5 results for Scotland, Psychological distress is identified using the Kessler scale, Obesity is based on UK90 thresholds for sex and age, Regular smoking is more than 6 cigarettes a week.

**Table A8. Simulated effect size of scenarios per child in birth cohort**

| Outcome                                                      | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4 |
|--------------------------------------------------------------|------------|------------|------------|------------|
| <b>Adverse outcomes (Reduction in probability by age 17)</b> |            |            |            |            |
| Poor GCSEs                                                   | 0.02       | 0.05       | 0.05       | 0.11       |
| Psychological distress                                       | 0.0        | 0.01       | 0.01       | 0.03       |
| Obesity                                                      | 0.0        | 0.01       | 0.01       | 0.04       |

| Outcome                                                              | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4 |
|----------------------------------------------------------------------|------------|------------|------------|------------|
| <b>Adverse outcomes (Reduction in probability by age 17)</b>         |            |            |            |            |
| Regular smoker                                                       | 0.0        | 0.02       | 0.02       | 0.03       |
| Poor health                                                          | 0.01       | 0.01       | 0.02       | 0.03       |
| Public cost savings (Cost savings between ages 3 and 17, in 2023 £s) |            |            |            |            |
| Total savings                                                        | 281        | 1,452      | 1,445      | 3,283      |
| By source                                                            |            |            |            |            |
| Any hospitalisation                                                  | −6         | −4         | 43         | 181        |
| Disability                                                           | 10         | 65         | 81         | 221        |
| Conduct disorder                                                     | 36         | 111        | 120        | 254        |
| Special education needs                                              | 153        | 1,102      | 1,015      | 2,316      |
| Persistent truancy                                                   | 2          | 9          | 8          | 19         |
| Permanent exclusion                                                  | 86         | 169        | 178        | 292        |
| Wellbeing improvement (WELLBYs gained between ages 3 and 17)         |            |            |            |            |
| WELLBYs                                                              | 0.4        | 1.05       | 1.31       | 3.21       |
| Value (in 2023 £s)                                                   | 5,253      | 13,618     | 17,059     | 41,782     |

Costs and wellbeing values are discounted at 3.5% per year calculated at birth. Tables with baseline levels and unadjusted figures can be found in the online appendix. Income increase scenarios - Scenario 1 - Poorest to second poorest, Scenario 2 - Poorest two to middle, Scenario 3 - Poorest to richest, Scenario 4 - All to richest Negative values in the table are increases in cases or public costs. Poor GCSEs also includes N5 results for Scotland, Psychological distress is identified using the Kessler scale, Obesity is based on UK90 thresholds for sex and age, Regular smoking is more than 6 cigarettes a week.

**Table A9.** Simulated effect size of scenarios per recipient in birth cohort

| Outcome                                                              | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4 |
|----------------------------------------------------------------------|------------|------------|------------|------------|
| <b>Adverse outcomes (Reduction in probability by age 17)</b>         |            |            |            |            |
| Poor GCSEs                                                           | 0.08       | 0.13       | 0.26       | 0.14       |
| Psychological distress                                               | 0.01       | 0.02       | 0.06       | 0.04       |
| Obesity                                                              | −0.01      | 0.02       | 0.07       | 0.06       |
| Regular smoker                                                       | 0.03       | 0.05       | 0.08       | 0.04       |
| Poor health                                                          | 0.03       | 0.04       | 0.08       | 0.04       |
| Public cost savings (Cost savings between ages 3 and 17, in 2023 £s) |            |            |            |            |
| Total savings                                                        | 1,482      | 3,762      | 7,618      | 4,128      |
| By source                                                            |            |            |            |            |
| Any hospitalisation                                                  | −33        | −10        | 229        | 227        |
| Disability                                                           | 53         | 169        | 429        | 278        |
| Conduct disorder                                                     | 192        | 288        | 633        | 320        |
| Special education needs                                              | 806        | 2,854      | 5,350      | 2,912      |
| Persistent truancy                                                   | 8          | 22         | 42         | 24         |
| Permanent exclusion                                                  | 455        | 438        | 936        | 367        |
| Wellbeing improvement (WELLBYs gained between ages 3 and 17)         |            |            |            |            |
| WELLBYs                                                              | 2.13       | 2.71       | 6.92       | 4.04       |
| Value (in 2023 £s)                                                   | 27,682     | 35,274     | 89,903     | 52,537     |

Costs and wellbeing values are discounted at 3.5% per year calculated at birth. Tables with baseline levels and unadjusted figures can be found in the online appendix. Income increase scenarios - Scenario 1 - Poorest to second poorest, Scenario 2 - Poorest two to middle, Scenario 3 - Poorest to richest, Scenario 4 - All to richest Negative values in the table are increases in cases or public costs. Poor GCSEs also includes N5 results for Scotland, Psychological distress is identified using the Kessler scale, Obesity is based on UK90 thresholds for sex and age, Regular smoking is more than 6 cigarettes a week.

**Table A10.** Effects of increasing income by \$1000 (£676) in **Cooper and Stewart (2021)** compared to three LifeSim models

| Cooper and Stewart (2021)                                                 |                    |        | LifeSim Childhood      |                 |                    |            |
|---------------------------------------------------------------------------|--------------------|--------|------------------------|-----------------|--------------------|------------|
| Paper                                                                     | Outcome            | Effect | Outcome                | Preferred Model | Extra Conservative | Unadjusted |
| Blau (1999) ( <i>All families in US</i> )                                 |                    |        |                        |                 |                    |            |
|                                                                           | PIAT Maths         | 0.01   | NFER Progress in maths | 0.02            | 0.01               | 0.02       |
|                                                                           | PIAT Reading       | 0.01   | BAS Word reading       | 0.02            | 0.01               | 0.03       |
|                                                                           | PPVT               | 0.01   | BAS Naming vocabulary  | 0.01            | 0.01               | 0.03       |
| Votruba-Drzal (2006) ( <i>All families in US</i> )                        |                    |        |                        |                 |                    |            |
|                                                                           | PIAT Maths         | 0.02   | NFER Progress in maths | 0.02            | 0.01               | 0.02       |
|                                                                           | PIAT Reading       | 0.02   | BAS Word reading       | 0.02            | 0.01               | 0.03       |
| Fernald et al. (2008) ( <i>Poor households in Mexico</i> )                |                    |        |                        |                 |                    |            |
|                                                                           | PPVT               | 0.21   | BAS Naming vocabulary  | 0.03            | 0.02               | 0.05       |
| Milligan and Stabile (2011) ( <i>Low education households in Canada</i> ) |                    |        |                        |                 |                    |            |
|                                                                           | PIAT Maths         | 0.07   | NFER Progress in maths | 0.03            | 0.03               | 0.05       |
|                                                                           | Hyperactivity      | 0.07   | SDQ Hyperactivity      | 0.0             | 0.0                | 0.0        |
|                                                                           | Conduct disorder   | 0.10   | SDQ Conduct problems   | 0.03            | 0.02               | 0.05       |
| Dahl and Lochner (2012) ( <i>Poor families in US</i> )                    |                    |        |                        |                 |                    |            |
|                                                                           | Maths              | 0.21   | NFER Progress in maths | 0.03            | 0.03               | 0.05       |
|                                                                           | Reading            | 0.21   | BAS Word reading       | 0.03            | 0.03               | 0.05       |
| Gennetian and Miller (2002) ( <i>Poor families in Minnesota</i> )         |                    |        |                        |                 |                    |            |
|                                                                           | BPI Internalising  | 0.12   | SDQ Internalising      | 0.03            | 0.02               | 0.05       |
|                                                                           | BPI Externalising  | 0.11   | SDQ Externalising      | 0.02            | 0.01               | 0.04       |
| Dearing et al. (2006) ( <i>Poor households in US</i> )                    |                    |        |                        |                 |                    |            |
|                                                                           | CBCL Internalising | 0.02   | SDQ Internalising      | 0.03            | 0.02               | 0.05       |
|                                                                           | CBCL Externalising | 0.03   | SDQ Externalising      | 0.01            | 0.01               | 0.03       |
| Zachrisson and Dearing (2015) ( <i>All families in Norway</i> )           |                    |        |                        |                 |                    |            |
|                                                                           | CBCL Internalising | 0.02   | SDQ Internalising      | 0.01            | 0.01               | 0.02       |

## Cooper and Stewart (2021)

## LifeSim Childhood

Effect sizes are based on standard deviation increases in outcome for a \$1000 increase in income. The most similar available test, based on concept measured and age at measurement, in the MCS is used for comparison with those presented in Cooper and Stewart (2021). The age range for the LifeSim estimates are based on the sweep age rather than actual age at time of testing. All MCS measures, except the SDQ scores, are age and ability adjusted measures that are standardised within sample. PIAT - Peabody individual achievement test, NFER - National foundation for educational research, PPVT - Peabody picture vocabulary test, BAS - British ability scales, SDQ - Strength and difficulties questionnaire, BPI - Basic personality inventory, and CBCL - Child behaviour checklist. Ages in MCS correspond to MCS sweeps and not age at observation (age 3 - sweep 2, age 5 - sweep 3 and so on until age 17 - sweep 7).

**Table A11.** Coefficients - marginal effects on life satisfaction (0-10)

| Life satisfaction age 17 |                   |                          |                   |
|--------------------------|-------------------|--------------------------|-------------------|
| Intercept                | 8.456<br>(0.114)  | Mixed                    | 0.041<br>(0.085)  |
| Poorest fifth            | -0.65<br>(0.058)  | Indian                   | 0.172<br>(0.096)  |
| Second poorest fifth     | -0.451<br>(0.048) | Pakistani or Bangladeshi | 0.204<br>(0.07)   |
| Middle fifth             | -0.273<br>(0.042) | Black                    | 0.158<br>(0.092)  |
| Second richest fifth     | -0.198<br>(0.039) | Other                    | 0.118<br>(0.132)  |
| Wales                    | -0.033<br>(0.065) | Smoked during pregnancy  | -0.198<br>(0.034) |
| Scotland                 | 0.014<br>(0.062)  | Age of mother at birth   | 0.004<br>(0.003)  |
| Northern Ireland         | 0.219<br>(0.071)  | No qualifications        | -0.237<br>(0.07)  |
| North East               | 0<br>(0.08)       | NVQ level 1              | -0.233<br>(0.072) |
| North West               | 0.007<br>(0.058)  | NVQ level 2              | -0.103<br>(0.057) |
| Yorkshire and the Humber | -0.051<br>(0.065) | NVQ level 3              | -0.108<br>(0.064) |
| East Midlands            | -0.009<br>(0.067) | NVQ level 4              | -0.02<br>(0.053)  |
| West Midlands            | -0.125<br>(0.059) | Disability in household  | -0.516<br>(0.082) |
| East of England          | -0.107<br>(0.057) | Single parent            | -0.056<br>(0.046) |
| South East               | -0.048<br>(0.052) |                          |                   |
| South West               | -0.044<br>(0.063) |                          |                   |

Life satisfaction is a linear transformation of SDQ emotional symptoms scale. The estimates presented are from an OLS regressions using the preferred specification on Life satisfaction on a 0 to 10 scale.

**Table A12.** Marginal effect on life satisfaction of a one unit change in log household income in early childhood

| Dependent variable                      | Age 17         |       | Age 14         |       | Age 11         |       |
|-----------------------------------------|----------------|-------|----------------|-------|----------------|-------|
|                                         | Coef<br>(s.e.) | Ratio | Coef<br>(s.e.) | Ratio | Coef<br>(s.e.) | Ratio |
| Parent reported SDQ emotional symptoms  | 0.39<br>(0.05) | 1     | 0.40<br>(0.04) | 1     | 0.34<br>(0.04) | 1     |
| Parent reported SDQ internalising score | 0.50<br>(0.05) | 1.28  |                |       |                |       |
| Self reported SDQ emotional symptoms    | 0.32<br>(0.07) | 0.82  |                |       |                |       |

| Dependent variable    | Age 17         |       | Age 14         |       | Age 11         |       |
|-----------------------|----------------|-------|----------------|-------|----------------|-------|
|                       | Coef<br>(s.e.) | Ratio | Coef<br>(s.e.) | Ratio | Coef<br>(s.e.) | Ratio |
| Self reported SWEMWBS | 0.21<br>(0.04) | 0.54  |                |       |                |       |
| Self reported LS1–7   |                |       | 0.23<br>(0.04) | 0.57  | 0.18<br>(0.04) | 0.53  |

Coefficients and standard errors are calculated using the confounder list from the preferred model. Ratio is the ratio of the coefficient of our main wellbeing to other available measures in the MCS.

**Table A13.** Effect of Scenario 1 on early childhood outcomes based on income at 9 months and incomes 9-months to age 5

| Outcome                                         | Income<br>Ages 0 - 5 | Income<br>at 9 months |
|-------------------------------------------------|----------------------|-----------------------|
| Hospitalisation<br>Age 3                        | 0.01<br>(0.02)       | 0.01<br>(0.02)        |
| Hospitalisation<br>Age 5                        | 0.01<br>(0.02)       | 0.01<br>(0.02)        |
| Disability<br>Age 3                             | 0.00<br>(0.02)       | 0.01<br>(0.02)        |
| Disability<br>Age 5                             | 0.00<br>(0.02)       | 0.01<br>(0.02)        |
| Parent reported SDQ<br>emotional symptoms Age 3 | -0.28<br>(0.10)      | -0.23<br>(0.10)       |
| Parent reported SDQ<br>emotional symptoms Age 5 | -0.20<br>(0.14)      | -0.09<br>(0.13)       |

Coefficients and standard errors are calculated using the confounder list from the preferred model.